Exact complex coherence in codimension two

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Abstract

We determine the minimum coherence of N unit vectors in $\mathbb{C}^{N-2}$ for every integer $N \geq 180$. The optimum is given by an explicit four-block construction with balanced integer multiplicities. The same optimum holds among all complex $N \times N$ matrices with diagonal entries one and rank at most $N-2$. Every extremizer is equivalent to the balanced four-block Gram matrix. The proof uses kernel duality to reduce the matrix problem to a geometric comparison in $\mathbb{C}^2$. Any competing matrix whose coherence is no larger than that of the construction admits a normalized two-dimensional left-kernel representation with every column near one of four directions. A strict local comparison then determines all equality cases. Rational arithmetic certificates establish the finite inequalities needed for the range $N \geq 180$. For $4 \nmid N$, the exact value improves the classical codimension-two bound by a positive term of order N^{-3}.

Contents

1 The optimization problem and the main result **3**

2 Relation to earlier work **4**

3 The attaining vectors and their stress **5**

4 From arbitrary matrices to the unique equality class **7**
4.1 The quantitative comparison used in the proof 7
4.2 Completion of the proof and equality cases 9

5 Further questions and verification **10**

6 Algebra of the four-weight construction **10**
6.1 The candidate domain needed at the endpoint 14
6.2 Exact projector and simultaneous polar estimates 14

7 Localization of every column at the endpoint 180 **17**
7.1 Deficit identity and estimates before partitioning 17
7.2 Simultaneous anchors and coverage 18
7.3 Counts before fine localization 19
7.4 Shared-budget forward refinement 20
7.5 Fine polar and spinor interface 21
7.6 Continuous range and zero budgets 22

8 Local comparison **22**
8.1 Inputs, gauge, and the normal decomposition 23
8.2 The reference tensor and an endpoint perturbation bound 24
8.3 Mean curvature in physical, mass, and stress coordinates 26
8.4 The intrinsic residual bound 28

8.5 The normal component and its finite budget . . . 30
8.6 Angular transport and the shared radial–tangent budget . . . 31
8.7 The pair remainder and physical fiber bounds . . . 32
8.8 Row estimates coupled to intrinsic residual curvature . . . 33
8.9 Complete scalar bounds and four-cluster summation . . . 34
8.10 A signed remainder bound . . . 36
8.11 Exact two-pair fibers and row cancellation . . . 38
8.12 Propagation and equality . . . 39

9 Exact certificates and the computational proof 41
9.1 Arithmetic and acceptance rules . . . 41
9.2 Global and continuous-parameter certificates . . . 42
9.3 Closed budget covers and local comparisons . . . 42
9.4 The worst $C = 3$ region . . . 44
9.5 Propagation to all integers in the stated range . . . 44
9.6 Certificates and arithmetic verification . . . 45

Acknowledgments 46

1 The optimization problem and the main result

The coherence of a family of unit vectors is the largest modulus of an inner product between distinct vectors. Minimizing coherence is a packing problem for complex lines: it asks how well a prescribed number of lines can be separated in a given dimension. All inner products are conjugate-linear in the first variable: $\langle v, w\rangle = v^* w$. For integers $1 \le n < N$ define

$$\mu_{\mathbb{C}}(n, N) = \min_{\substack{v_1,\dots,v_N \in \mathbb{C}^n \\ \|v_i\|=1}} \max_{i \neq j} |v_i^* v_j|.$$

The minimum exists: the product of N unit spheres is compact and the maximum of the finitely many continuous inner-product moduli is continuous.

We study the case $n = N - 2$. The associated Gram matrices have a kernel of dimension at least two; a two-dimensional subspace of that kernel provides the geometric representation used below. When $4 \mid N$, the classical attaining construction corresponds to equal replication of four equiangular lines in this representation [1, 8]. For the remaining cardinalities, equal replication is impossible. We construct a candidate using balanced integer multiplicities and deform the four-line configuration accordingly. We prove that this candidate is globally optimal among all vector families for every $N \ge 180$, and determine all equality cases.

The kernel argument also applies to matrices that need not initially be Gram matrices. We therefore consider the two matrix problems

$$a_N = \min_{\substack{A \in \mathbb{C}^{N\times N} \\ A_{ii}=1,\ \operatorname{rank} A \le N-2}} \max_{i \neq j} |A_{ij}|, \qquad b_N = \inf_{\substack{A \in \mathbb{C}^{N\times N} \\ A_{ii}=1,\ \operatorname{rank} A = N-2}} \max_{i \neq j} |A_{ij}|.$$

For $N \ge 3$ the first minimum also exists. The all-ones matrix is feasible with objective one, so we can restrict to the compact set with all off-diagonal moduli at most one. Diagonal and rank-at-most conditions are closed. Moreover $a_N > 0$, because a diagonal-one matrix with zero off-diagonal entries has rank N. No compactness of the exact-rank set is assumed; attainment of b_N will follow from the construction.

Set $\alpha = (1 + \sqrt{3})/4$. A four-tuple $\boldsymbol{m} = (m_1, m_2, m_3, m_4)$ of positive integers with sum N is *balanced* if its entries differ by at most one. Such a tuple is unique up to permutation. Put $p_a = m_a/N$ and denote by $\lambda_{\boldsymbol{p}}$ the root specified in the following theorem.

Theorem 1.1. *For every integer $N \ge 180$ and every balanced four-tuple $\boldsymbol{m}$ of positive integers with sum N, the equation*

$$2\lambda = \sum_{a=1}^{4} \frac{p_a(1 - 3p_a)}{\lambda - 2p_a} \tag{1.1}$$

has a unique root in $(2/3, \alpha]$, and

$$\mu_{\mathbb{C}}(N-2, N) = a_N = b_N = \frac{1}{N\lambda_{\boldsymbol{p}} - 1}. \tag{1.2}$$

The exact-rank infimum is attained. Every minimizing matrix in the rank-at-most problem is, up to index permutation, diagonal unitary switching and entrywise conjugation, the balanced four-block Gram matrix in Proposition 3.1. In particular it is positive semidefinite, has rank $N-2$, and all off-diagonal entries have the same modulus. Vector realizations differ additionally by a common unitary isometry.

For $4 \mid N$ the value is $1/(\alpha N - 1)$, the known divisible case. For $N \equiv 2 \pmod 4$ the formula becomes

$$\mu_{\mathbb{C}}(N-2, N) = \frac{4}{N + \sqrt{3N^2 - 8} - 4}. \tag{1.3}$$

Corollary 1.2. *For each fixed $r \in \{0,1,2,3\}$, as $N \longrightarrow \infty$ through $N \equiv r \pmod 4$,*

$$\mu_{\mathbb{C}}(N-2,N) = \frac{1}{\alpha N - 1} + \frac{r(4-r)}{4\sqrt{3}\,\alpha^2}\, N^{-3} + O(N^{-4}).$$

For $r \neq 0$ the correction is positive.

Proof. Write $p_a = 1/4 + \epsilon_a$, $\sum_a \epsilon_a = 0$ and $\sum_a \epsilon_a^2 = r(4-r)/(4N^2)$. Define $\Phi_{\boldsymbol{p}}(\lambda) = 4[2\lambda - \sum_a p_a(1-3p_a)/(\lambda - 2p_a)]$ and put $h = \alpha - 1/2$. Polynomial division gives

$$\Phi_{\boldsymbol{p}}(\alpha) = 4\sum_a \frac{\epsilon_a^2}{h - 2\epsilon_a}, \qquad \Phi'_{(1/4,\ldots,1/4)}(\alpha) = \frac{4\sqrt{3}}{h}.$$

The derivative is positive near uniform weights; Taylor's formula gives $\alpha - \lambda_{\boldsymbol{p}} = \sum_a \epsilon_a^2/\sqrt{3} + O(N^{-3})$. Substitution in (1.2) proves the assertion. □

The correction in Corollary 1.2 measures the cost of the integer multiplicities. It vanishes when equal replication is possible and is positive in each of the other residue classes.

The proof begins with a trace inequality relating the coherence of a matrix to an energy associated with a two-dimensional subspace of its left kernel. A matrix that matches or improves on the construction must have kernel energy close to the classical upper bound. We use the resulting shared deficit bounds to place every kernel column in one of four clusters and to restrict their integer counts. A local comparison then controls both motion of the cluster centers and variation within each cluster, including the complex phases. Its strict inequality forces an extremizer to coincide with the corresponding four-block candidate. Integer balancing selects the optimal counts, and equality in the trace inequality recovers the full matrix.

The explicit construction and matrix argument are given in Sections 3 and 4. The quantitative localization and local comparison use rational inequalities on closed domains, whose certificates are described in Section 9. The threshold 180 is sufficient; its minimality remains open.

2 Relation to earlier work

Bukh and Cox [1, Theorems 2(b) and 3(b)] proved a bound for general matrices. For a complex matrix A of order N with diagonal one and rank at most $N-k$, it gives

$$\max_{i\neq j} |A_{ij}| \geq \frac{1}{\alpha_k N - 1}, \qquad \alpha_k = \frac{1 + (k-1)\sqrt{k+1}}{k^2}. \tag{2.1}$$

Although their extremal quantity is defined using exact rank, the proof of their Theorem 10 already permits rank at most $N-k$. Equality in (2.1) requires a maximal equiangular tight frame in $\mathbb{C}^k$ and $k^2 \mid N$; uniform replication supplies the matching construction. In codimension two this is the tetrahedral configuration of four complex lines. Consequently, the exact value $((1+\sqrt{3})N/4 - 1)^{-1}$ when $4 \mid N$ is an established result.

Their corrigendum [2] specifies the complex equality case: equal mass is assigned to each entire unit-phase orbit, rather than to a pair of antipodal representatives. This correction preserves the numerical bound and its divisibility condition. We use the corrected projective formulation throughout.

Magsino, Mixon and Parshall [11, Lemma 1 and Theorem 4] give a Delsarte-style proof through an auxiliary polynomial and tight-frame duality. Glazyrin [8, Theorem 7] develops a general matrix duality for conjugate energies. His replicated-code construction and equality analysis [8, Theorem 10 and the subsequent discussion] also show that an extremizing general matrix is a Gram matrix, with uniqueness inherited from the maximal projective simplex. In

particular, the divisible codimension-two case already has this stronger equality description. We use this duality, the auxiliary-polynomial estimate, phase alignment at equality, and the tetrahedral model in the proof below.

Two closely related optimization problems help explain the dual functional. Cahill and Casazza [3, Proposition 2.1 and Theorems 2.2, 2.4] study total coherence of Parseval frames, its invariance under Naimark complementation, and restrictions on maximizing frames. For columns u_i of a matrix $U \in \mathbb{C}^{2\times N}$ with $UU^* = I_2$, write

$$\mathrm{TC}(U) = \sum_{i\neq j} |u_i^* u_j|.$$

The quasimaximal relative projection constant of Foucart and Skrzypek [7, Section 2] is, in this notation,

$$q_{\mathbb{C}}(2,N) = \frac{\max_{UU^*=I_2} \mathrm{TC}(U) + 2}{N}.$$

Their maximal relative projection constant additionally optimizes independent nonnegative weights t_i, with $\sum_i t_i^2 = 1$, in $\sum_{i,j} t_i t_j |u_i^* u_j|$. These distinctions matter when transferring an exact value or an equality statement to the coherence problem.

Kobos [9, Theorem 4.2] proves a stability theorem: a sequence approaching the maximal ETF bound admits, after passing to a subsequence, clustering around maximal ETF directions in a weighted mean-square sense; an exceptional set may have vanishing total weight. His analysis also treats phases and residuals. To obtain the finite conclusion here, we need bounds that place every column in a controlled neighborhood at each cardinality under consideration. The subsequent local comparison must eliminate all within-cluster residuals. These are the quantitative steps developed in this paper.

Deręgowska and Lewandowska [5, Theorem 3.6] determine relative projection constants at replicated maximal-ETF cardinalities. Kumar, Mohar, Mojallal and Pragada [10, Theorem 1.3] establish eventual stabilization of maximal relative projection constants in the real setting. The codimension-two theorem of Deręgowska and Lewandowska [6, Theorem 2.4] instead controls a real Banach–Mazur distance through a small projection constant. These results concern different objectives from the finite complex off-diagonal minimax problem in Theorem 1.1.

Theorem 1.1 extends the exact value and complete equality classification to the nondivisible cardinalities in the range $N \geq 180$, within the full class of complex unit-diagonal matrices of rank at most $N-2$. It identifies the balanced integer multiplicities, the corresponding deformation of the four-line configuration, and all extremizers. Quantitative local rigidity and computer-assisted existence arguments also occur in projective coding theory, for example in [4]. Here the local comparison is combined with control of every kernel column to obtain a global minimax theorem uniformly over the stated cardinalities.

3 The attaining vectors and their stress

We give the construction for every *admissible* profile $\boldsymbol{p} = (p_1, \dots, p_4)$, meaning $0 < p_a < 1/3$ and $\sum p_a = 1$. Let $\lambda = \lambda_{\boldsymbol{p}}$ be the root of (1.1), and put

$$d_a = \lambda - p_a, \qquad w_a = p_a/d_a, \qquad X_{aa} = 1, \quad X_{ab} = 2w_a w_b - 1 \quad (a \neq b). \tag{3.1}$$

The scalar equation has a unique root in $(2/3, \alpha]$, and X is a real positive semidefinite matrix of rank three. The algebraic proofs, including strict integer balancing, are in Section 6. Realize X by four unit Bloch vectors $x_a \in \mathbb{R}^3$ and choose unit spinors $u_a \in \mathbb{C}^2$ with $|u_a^* u_b|^2 = (1 + x_a \cdot x_b)/2$.

Proposition 3.1 (Explicit vector factor)**.** *Let $m_a = Np_a$ be positive integers. Form the 2×4 matrix $W = [\sqrt{d_1}u_1 \ \cdots \ \sqrt{d_4}u_4]$ and put $B = W^*W$, $A_p = \lambda I_4 - B$. Let J be the $N \times 4$ block*

isometry, with entry $1/\sqrt{m_a}$ *in column* a *on block* a*. If the columns of* $R \in \mathbb{C}^{N\times(N-4)}$ *form an orthonormal basis of* $\operatorname{range}(J)^{\perp}$*, then*

$$V_p = \frac{1}{\sqrt{N\lambda - 1}} \begin{pmatrix} \sqrt{N\lambda}\, R^* \\ \sqrt{N}\, W J^* \end{pmatrix} \in \mathbb{C}^{(N-2)\times N} \tag{3.2}$$

has unit columns, rank $N-2$*, and coherence* $(N\lambda-1)^{-1}$*. Its Gram matrix is*

$$S_p = NJA_pJ^*, \qquad G_p = V_p^*V_p = \frac{N\lambda I_N - S_p}{N\lambda - 1}. \tag{3.3}$$

For a fixed ordered profile, the Gram matrix G_p *is unique up to diagonal unitary conjugation and entrywise conjugation; permuting the profile permutes its blocks.*

Proof. The Bloch identity gives $B_{aa} = \lambda - p_a$ and $|B_{ab}| = \sqrt{p_a p_b}$ for $a \neq b$. The matrix B is positive semidefinite of rank two. Thus A_p has top eigenvalue λ with multiplicity exactly two. Every entry of S_p has modulus one, and its diagonal entries equal one. Consequently $N\lambda I - S_p$ is positive semidefinite with exactly two zero eigenvalues and constant diagonal $N\lambda - 1$. Moreover

$$N\lambda I - S_p = N\lambda(I - JJ^*) + NJBJ^*.$$

Using $RR^* = I - JJ^*$ proves (3.2) and all the claimed rank and modulus assertions. Orthogonal realizations of the rank-three real Gram matrix X differ by a rotation, or a rotation followed by a reflection. On spinors these give a common unitary, or a common unitary followed by conjugation; individual lift phases give diagonal unitary conjugation. The choice of R changes only the common unitary factor of the vector realization. □

To compare the construction with arbitrary matrices, we associate a positive 2×2 matrix K to its kernel. We call K the candidate stress. The trace bound in Section 4 will convert an upper bound for its kernel functional Γ_K into a coherence lower bound. Set

$$q_a = \frac{(1-\lambda-p_a)(\lambda-p_a)}{2(1-\lambda)(\lambda-2p_a)}, \quad \ell_a = \frac{2q_a}{\lambda - p_a}, \quad L = \operatorname{diag}(\ell_a), \quad T = L - LBL.$$

Choose $U \in \mathbb{C}^{4\times 2}$ with $U^*U = I$ and $BU = 0$. Writing $Y = [y_1 \ \cdots \ y_N] \in \mathbb{C}^{2\times N}$, define

$$Z = \sqrt{N/2}\, U^*J^*, \qquad K = \frac{2U^*TU}{\operatorname{tr} T}, \qquad \Gamma_K(Y) = \sum_{i,j} |y_i^* K y_j|. \tag{3.4}$$

The identities $q_a > 0$, $\sum q_a = 1$, $Xq = 0$ imply $WLW^* = I$. Hence $T \succeq 0$ has rank two and range $\ker B$, so $K \succ 0$ and $\operatorname{tr} K = 2$. Section 6 verifies that

$$ZZ^* = \frac{N}{2} I, \quad ZS_p = N\lambda Z, \quad \operatorname{phase}(Z^*KZ) = S_p, \quad \Gamma_K(Z) = N^2\lambda. \tag{3.5}$$

In particular, $ZG_p = 0$, so Z is the normalized left-kernel frame of the candidate Gram matrix, with columns constant on each block. All overlaps in the phase identity are nonzero, including those within a block. Holding K fixed, Z is stationary for $\Gamma_K(Y) - N\lambda \operatorname{tr}(KYY^*)$.

Finally, strict convexity of $p \mapsto (\lambda - 2p)^{-1}$ in the scalar equation shows that transferring one count from a larger block to a block smaller by at least two strictly increases λ_p. Thus

$$\lambda_p \leq \lambda_{\mathrm{bal}}, \qquad \text{with equality exactly for balanced integer counts.} \tag{3.6}$$

After localization, this comparison selects the balanced profile from the possible cluster counts.

4 From arbitrary matrices to the unique equality class

The left kernel of a matrix in our rank-constrained class supplies the comparison functional. A positive stress gives a trace bound whose equality condition determines the matrix entries.

Lemma 4.1 (Left-kernel trace bound)**.** *Let $A \in \mathbb{C}^{N\times N}$ have diagonal one, rank at most $N-2$, and coherence $\mu = \max_{i\neq j}|A_{ij}|$. Choose $Y \in \mathbb{C}^{2\times N}$ with rows in the left kernel of A and normalization $YY^* = NI_2/2$, so that $YA = 0$. For every $K \succ 0$ with $\operatorname{tr} K = 2$,*

$$N(1+1/\mu) \le \Gamma_K(Y). \tag{4.1}$$

If equality holds and all off-diagonal entries of $H = Y^*KY$ are nonzero, then $A_{ij} = -\mu H_{ij}/|H_{ij}|$ for every $i \neq j$.

Proof. Take two orthonormal columns Q in $\ker A^*$ and put $Y = \sqrt{N/2}\,Q^*$. Then $HA = 0$, $H \succeq 0$, and $\operatorname{tr} H = N$. Also $\mu > 0$, since the identity has rank N. Expanding $\operatorname{tr}(HA) = 0$, separating the diagonal terms, and taking real parts gives

$$N = -\sum_{i\neq j}\operatorname{Re}(H_{ij}A_{ji}) \le \mu\sum_{i\neq j}|H_{ij}| = \mu(\Gamma_K(Y) - N).$$

If equality holds, every ordered term with $H_{ij} \neq 0$ attains both its modulus and real-part bounds. Thus $A_{ji} = -\mu\overline{H_{ij}}/|H_{ij}|$, which is the stated formula after exchanging the indices. □

4.1 The quantitative comparison used in the proof

For Y as in Lemma 4.1, write $y_i = r_iu_i$ and let $x_i \in S^2$ be the Bloch vector of u_i. If $r_i = 0$, choose u_i arbitrarily at this stage. Put

$$a = \frac{2\sqrt{3}-1}{8}, \quad k = \frac{6\sqrt{3}-9}{4}, \quad L = \sum r_i, \qquad g(t) = (1-\sqrt{t})(3/4 + k\sqrt{t})(\sqrt{t} - 1/\sqrt{3})^2.$$

The radial, moment and angular deficits are

$$u = a(N^2 - L^2), \qquad v = \frac{k}{4}\left\|\sum_i r_ix_ix_i^T - \frac{L}{3}I\right\|_F^2, \qquad w = \sum_{i,j} r_ir_j g((1 + x_i\cdot x_j)/2). \tag{4.2}$$

They are nonnegative and satisfy $u + v + w \le \alpha N^2 - \Gamma_I(Y)$, by the exact polynomial identity in Section 7. Applying Lemma 4.1 with $K = I$ to a competitor with $\mu \le U_N = (N\lambda_{\mathrm{bal}} - 1)^{-1}$ gives

$$u + v + w \le M, \qquad M = \begin{cases} 1/2, & N \text{ odd}, \\ 11/19, & N \equiv 2 \pmod 4. \end{cases} \tag{4.3}$$

The scalar candidate estimates establishing these budgets are included in Section 6.

Proposition 4.2 (Localization of every column)**.** *Suppose $N \ge 180$, $4 \nmid N$, $YY^* = NI_2/2$, and $\Gamma_I(Y) \ge N^2\lambda_{\mathrm{bal}}$. There is a partition of all N columns into four nonempty groups with positive counts m_a. Set $\delta_a = m_a - N/4$ and $C = \sum\delta_a^2$. Up to permutation and simultaneous sign reversal, its possible profiles are*

Residue	C	$(\delta_1,\delta_2,\delta_3,\delta_4)$
N *odd*	3/4	$(-1,-1,-1,3)/4$
N *odd*	11/4	$(-5,-1,3,3)/4$
$N \equiv 2 \pmod 4$	1	$(-1,-1,1,1)/2$
$N \equiv 2 \pmod 4$	3	$(-3,1,1,1)/2$

On this same partition let $\bar{x}_a$ be the group means and $S = \sum_a \sum_{i\in a} \|x_i - \bar{x}_a\|^2$. Then $S \le 31E/N$, where $E = \sum_{i,j} g((1 + x_i \cdot x_j)/2)$. The simultaneous total and pointwise estimates (7.14)–(7.15) hold in one common preliminary alignment to the repeated tetrahedral frame $F = (f_{a(i)})_i$, where $a(i)$ is the group containing column i and the f_a are unit tetrahedral spinors.

Proof. Section 7 keeps the shared budget $u + v + w \le M$ through anchor selection and partitions every column. Its centered count estimate gives $C < 5$ in residue two and $C < 19/4$ in the odd residues, which imply exactly the listed profiles. The forward refinement on this same partition gives the scatter and simultaneous polar estimates. The finite inequalities used in these steps are specified in Section 9. □

The scales explain the role of the local step. The shared deficit is $O(1)$, uniformly in N, and the count variance C is bounded. Localization and the candidate estimates place both Y and the candidate Z with the same profile within $O(N^{-1/2})$ of F in the Frobenius norm. The triangle inequality and minimization over the allowed alignments therefore give $d = \|Y - Z\|_F = O(N^{-1/2})$ in the closest gauge. The separate transport of pointwise bounds to that gauge is proved in Section 8.6. The relative normal component is at most $d/\sqrt{2N} = O(N^{-1})$. The horizontal reference curvature has a negative gap bounded away from zero; candidate perturbations and the higher-order loss tend to zero. This explains why the local comparison becomes strict for large N. To obtain the explicit threshold 180, we bound both terms on the localized regions and certify the resulting inequalities by rational arithmetic.

Proposition 4.3 (Conditional local comparison)**.** *Fix one of the localized profiles, its candidate Z and its positive stress K. Put a competitor Y with $YY^* = NI_2/2$ in the closest unitary and individual-phase gauge to Z, allowing conjugation of the candidate, and set $d = \|Y - Z\|_F$. Suppose all validity conditions in Section 8, including the denominator, positivity, phase, overlap and closest-gauge transport conditions, hold throughout a closed region containing this competitor. Let c_m, c_r be valid lower bounds for the curvature gaps in the mean and residual directions, and J, L valid upper bounds for the mixed and pure normal terms, measured in the original Frobenius metric as in (4.5). Let $A_{\mathrm{rem}}, B_{\mathrm{rem}} \ge 0$ supply one bound for the full Taylor remainder using the common tangent fraction defined below. Define g_{normal} as the negative of the maximum of the quadratic upper bound in (4.5) over the normalized components, with the normal fraction restricted by the bound for this region. If*

$$g_{\mathrm{normal}} - \frac{A_{\mathrm{rem}} + \sqrt{A_{\mathrm{rem}}^2 + B_{\mathrm{rem}}^2}}{2} \ge \varepsilon > 0,$$

then

$$\Gamma_K(Y) \le N^2\lambda_p - \varepsilon N\|Y - Z\|_F^2. \tag{4.4}$$

Proof. The case $d = 0$ is immediate. For $d > 0$, define the Lagrangian $\mathcal{L}(Y) = \Gamma_K(Y) - N\lambda_p \operatorname{tr}(KYY^*)$ and its half-Hessian $Q(H) = \frac{1}{2}D^2\mathcal{L}(Z)[H, H]$, with K fixed and derivatives taken over the underlying real vector space. Put $\Delta = Y - Z$ and $R_n = (\Delta\Delta^*/N)Z$. The horizontal component $\Xi = \Delta + R_n$ splits orthogonally into cluster means H_{m} and zero-sum residuals H_{r}. Set $h = \|H_{\mathrm{m}}\|_F/d$, $r = \|H_{\mathrm{r}}\|_F/d$, and $n = \|R_n\|_F/d$. Under the stated validity conditions, the half-Hessian estimate is

$$\frac{Q(\Delta)}{Nd^2} \le -c_m h^2 - c_r r^2 + 2Jnh + Ln^2, \quad h^2 + r^2 + n^2 = 1, \quad n \le d/\sqrt{2N}. \tag{4.5}$$

The exact maximum of this quadratic upper bound over the allowed normal fraction, including both branches, is $-g_{\mathrm{normal}}$; see (8.24).

The remainder uses the distinct physical decomposition $y_i = (1 + t_i)z_i + \xi_i$, where $t_i \in \mathbb{R}$ and $z_i^*\xi_i = 0$. For the common tangent fraction $\theta = (\sum_i \|\xi_i\|^2)^{1/2}/d \in [0, 1]$,

its complete upper bound is $Nd^2[A_{\mathrm{rem}}\theta^2 + B_{\mathrm{rem}}\theta\sqrt{1-\theta^2}]$. Lemma 8.9 bounds this by $Nd^2(A_{\mathrm{rem}} + \sqrt{A_{\mathrm{rem}}^2 + B_{\mathrm{rem}}^2})/2$. The first variation of $\mathcal{L}$ vanishes at Z, and the constraint term $N\lambda_p \operatorname{tr}(KYY^*)$ has the same value at Y and Z because their covariances agree. Adding the quadratic and remainder bounds proves (4.4). Section 8 derives the bounds and their validity conditions; the next proposition certifies those conditions on the regions supplied by localization. □

Proposition 4.4 (Finite verification and propagation)**.** *The exact verification statement in Section 9.3 covers all closed budget cells of the four profiles. The local endpoints are* 180 *for* $C \in \{3/4, 11/4, 3\}$ *and* 182 *for* $C = 1$*. Each cell is either excluded by a strict counting inequality or lies in a closed region where all validity conditions of Proposition 4.3 hold and the quadratic gap exceeds the remainder bound by more than* $1/10000$*. For each fixed signed deviation vector* (δ_a)*, these inequalities remain valid for every larger integer* N *for which* $m_a = N/4 + \delta_a$ *are integers.*

Proof. The rational records and the conditions checked by the verifier are given in Section 9.3. Lemma 8.10 extends the verified inequalities by fixing valid endpoint coefficients and bounding the reciprocal parameter over the full subsequent interval. □

Corollary 4.5 (Localization and strict comparison)**.** *Under the hypotheses of Proposition 4.2, use its partition and the closest gauge to the candidate with the same profile. Then*

$$\Gamma_K(Y) \le N^2\lambda_p - \frac{N}{10000}\|Y - Z\|_F^2. \tag{4.6}$$

Proof. Combine Propositions 4.2, 4.3 and 4.4. Odd integers in the range start at 181; the first actual residue-two integer is 182. Thus the two different local endpoints leave no integer gap for nondivisible $N \ge 180$. □

4.2 Completion of the proof and equality cases

We now prove Theorem 1.1. For $4 \nmid N$, take any complex $N \times N$ matrix A with diagonal entries one, rank at most $N-2$, and coherence $\mu \le U_N$. Choose Y as in Lemma 4.1. The closest gauge is legitimate for a general matrix: replacing Y by UYD is accompanied by $A \mapsto D^*AD$. It preserves $YA = 0$, covariance, diagonal, rank and coherence. Select the stress K for this profile *after* this alignment. Lemma 4.1, Corollary 4.5 and strict balancing give

$$\begin{aligned} N(1 + 1/\mu) &\le \Gamma_K(Y) \\ &\le N^2\lambda_p - \frac{N}{10000}\|Y - Z\|_F^2 \\ &\le N^2\lambda_{\mathrm{bal}} \le N(1 + 1/\mu). \end{aligned}$$

Every inequality is an equality. Therefore $Y = Z$ in the chosen gauge, the counts are balanced, and $\mu = U_N$. All stress overlaps are nonzero, so the ordered equality condition in Lemma 4.1 forces

$$A = (1 + \mu)I - \mu S_p = G_p.$$

Undoing the allowed transformations gives exactly the equality class of the theorem.

For $4 \mid N$, the classical scalar bound is sharp. For completeness, the deficit identity at zero forces $r_i = 1$, vanishing moment defect, and $g((1 + x_i \cdot x_j)/2) = 0$ for every pair. Thus distinct Bloch directions have inner product $-1/3$. At most four such directions exist. Since $r_i = 1$, tightness gives $\sum x_i = 0$; together with the zero moment defect this gives $\sum l_i l_i^T = NI_4/4$, where $l_i = (1, \sqrt{3}x_i)^T/2$. The lifted directions therefore span $\mathbb{R}^4$, so there are exactly four. They form an orthonormal basis of $\mathbb{R}^4$, and tightness forces equal counts $N/4$. Hence the

kernel consists of equally repeated tetrahedral spinor lines. Equality in the ordered trace bound with $K = I$ again fixes $A = G_p$. This also proves the full matrix equality statement in the divisible case.

Proposition 3.1 supplies an attaining rank-exactly $N - 2$ Gram matrix. Since the rank-at-most problem is a relaxation of both other problems, all three optimal values coincide. Finally, two full-rank vector factors with the same Gram matrix differ by a common unitary on $\mathbb{C}^{N-2}$, completing the vector classification.

5 Further questions and verification

The theorem resolves the nondivisible cardinalities in an explicit sufficient range. Determining the least threshold for the balanced four-block construction remains open. In higher codimension, a related question is whether the extremizers at nondivisible cardinalities arise by deforming replicated maximal equiangular tight frames, when such frames exist. An extension would require both a localization theorem for the corresponding projective configuration and a strict comparison for its possible integer multiplicities.

Two features of the present argument may be useful for these questions. The common deficit budget couples radial and angular deviations during localization. With the candidate stress fixed, the kernel functional Γ_K is exactly quadratic under positive radial rescalings of the columns. The radial–tangential decomposition in the local comparison uses this property to control the remainder. These features reduce the loss in passing from a global bound to a finite neighborhood of the candidate.

The proof uses computer assistance to verify the finite inequalities in Sections 6–9. The supplementary package provides rational certificates, an arithmetic verifier that checks them without repeating the certificate search, and instructions for reproducing the verification. Section 9 specifies the verifier's scope and the analytic arguments that extend the finite checks to the stated range.

6 Algebra of the four-weight construction

Throughout, $\langle u, v\rangle = u^* v$, and a collection of N vectors is written as a matrix with N columns. Put $\alpha = (1 + \sqrt{3})/4$. A profile is admissible if $0 < p_a < 1/3$ for $1 \le a \le 4$ and $\sum_a p_a = 1$. Unsubscripted matrix norms are operator norms. We distinguish the kernel projector Π, the mass matrix D_{mass}, the Bloch Gram difference Δ_{B}, and the scalar candidate point bound C_{pt}. The local Hessian error bound in Section 8 is denoted E_{Hess}.

Lemma 6.1 (The scalar parameter)**.** *For every admissible profile there is exactly one* $\lambda_p \in (2/3, \alpha]$ *satisfying*

$$2\lambda_p = \sum_a \frac{p_a(1 - 3p_a)}{\lambda_p - 2p_a}. \tag{6.1}$$

Moreover $\lambda_p < 1 - \max_a p_a$*, and* $\lambda_p = \alpha$ *if and only if all four weights are* $1/4$*.*

Proof. For $\lambda \ge 2/3$ all denominators below are positive. Define

$$\begin{aligned} \Phi_p(\lambda) &= \lambda(3\lambda - 2) \sum_a \frac{1}{\lambda - 2p_a} - 2(2\lambda - 1) \\ &= 4\left[2\lambda - \sum_a \frac{p_a(1 - 3p_a)}{\lambda - 2p_a}\right]. \end{aligned} \tag{6.2}$$

Polynomial division and $\sum p_a = 1$ give the second identity. Thus

$$\Phi_p(2/3) = -2/3, \qquad \Phi_p'(\lambda) = 4\left[2 + \sum_a \frac{p_a(1-3p_a)}{(\lambda-2p_a)^2}\right] > 8.$$

Strict convexity gives $\sum_a (\alpha - 2p_a)^{-1} \geq 4/(\alpha - 1/2)$. Since $\alpha(3\alpha - 2) > 0$, substitution yields $\Phi_p(\alpha) \geq 0$, with equality precisely at the uniform profile. Continuity and strict monotonicity prove existence and uniqueness. For the extra bound put $p = \max p_a$ and $L = 1 - p > 2/3$. The other three weights sum to L, so their denominators $L - 2p_a$ sum to L. Therefore

$$\sum_a \frac{1}{L-2p_a} \geq \frac{1}{3L-2} + \frac{9}{L}, \qquad \Phi_p(L) \geq L(3L-2)\left[\frac{1}{3L-2} + \frac{9}{L}\right] - 2(2L-1) = 8(3L-2) > 0.$$

Strict monotonicity implies $\lambda_p < L$. □

Abbreviate $\lambda = \lambda_p$, and put

$$d_a = \lambda - p_a, \quad w_a = p_a/d_a, \quad X_{aa} = 1, \quad X_{ab} = 2w_a w_b - 1 \quad (a \neq b). \tag{6.3}$$

Here X is an auxiliary four-point real Gram matrix, distinct from the Bloch points of an arbitrary competitor.

Lemma 6.2 (The auxiliary Gram matrix and its positive kernel)**.** *The matrix X is positive semidefinite of rank three. Its kernel is spanned by the strictly positive vector*

$$q_a = \frac{(1-\lambda-p_a)(\lambda-p_a)}{2(1-\lambda)(\lambda-2p_a)}. \tag{6.4}$$

It has $\sum q_a = 1$, and $q_a < 1/2$ for every a.

Proof. We have $0 < w_a < 1$. Set $D_X = 2\,\mathrm{diag}(1 - w_a^2)$, so $X = D_X + 2ww^T - \mathbf{1}\mathbf{1}^T$. Write $A_s = \sum(1-w_a^2)^{-1}$, $B_s = \sum w_a/(1-w_a^2)$, and $R = \sum(\lambda - 2p_a)^{-1}$. Expanding the 2×2 determinant for this rank-two update gives

$$\frac{\det X}{\det D_X} = (A_s - 3)(1 - A_s/2) + B_s^2/2.$$

The identities

$$A_s - B_s = 4 - 1/\lambda, \qquad A_s + B_s = 2 + \lambda R/2$$

make $\det X = 0$ equivalent to $(3\lambda - 2)(2 + \lambda R/2) = 8\lambda - 5$, which is $\Phi_p(\lambda) = 0$. The matrix $D_X + 2ww^T$ is positive definite. On $\mathbf{1}^\perp$ its quadratic form remains unchanged after subtracting $\mathbf{1}\mathbf{1}^T$. Thus X has at least three strictly positive eigenvalues; its remaining eigenvalue is zero.

Every factor in (6.4) is positive by Lemma 6.1. The normalization and kernel identities can be checked without an eigenvalue computation. For arbitrary λ in the displayed domain, put $\beta = (2\lambda - 1)/(2(1-\lambda))$. Division in $\lambda - 2p_a$ gives

$$\sum_a q_a - 1 = \sum_a w_a q_a - \beta = -\frac{\Phi_p(\lambda)}{8(1-\lambda)}. \tag{6.5}$$

Also $2(1 - w_a^2)q_a = 1 - 2\beta w_a$. Consequently $(Xq)_a = (1 - 2w_a)\Phi_p(\lambda)/(8(1-\lambda))$. At the root, $\sum q_a = 1$ and $Xq = 0$. Realize X by unit vectors $x_a \in \mathbb{R}^3$. Then $\sum q_a x_a = 0$, so $q_a = \|\sum_{b \neq a} q_b x_b\| \leq 1 - q_a$. Equality would force the three other points to coincide at $-x_a$, contradicting rank three. Hence $q_a < 1/2$. □

We use the elementary Bloch representation $uu^* = (I + x \cdot \sigma)/2$ of a unit spinor $u \in \mathbb{C}^2$, where

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Multiplication and traces of these matrices give

$$|u^* v|^2 = (1 + x \cdot y)/2. \tag{6.6}$$

Conversely a spinor for $x \in S^2$ is

$$u(x) = \left(\sqrt{(1 + x_3)/2}, \frac{x_1 + i x_2}{\sqrt{2(1 + x_3)}} \right)^T$$

when $x_3 \neq -1$, and $(0, 1)^T$ at the south pole.

Proposition 6.3 (An explicit attaining configuration)**.** *Let m_a be positive integers with total N and admissible $p_a = m_a/N$. There are N unit vectors in $\mathbb{C}^{N-2}$ whose Gram matrix has rank exactly $N - 2$ and constant off-diagonal absolute value*

$$\mu_p = \frac{1}{N\lambda_p - 1}. \tag{6.7}$$

To construct them, realize X by real unit vectors x_a, choose corresponding unit spinors u_a, and let $W \in \mathbb{C}^{2\times 4}$ have columns $\sqrt{d_a} u_a$. Set

$$\begin{aligned} B = W^* W, \qquad & A_p = \lambda I_4 - B, \\ D_{\mathrm{mass}} = \mathrm{diag}(\sqrt{p_a}), \qquad & C = D_{\mathrm{mass}}^{-1} A_p D_{\mathrm{mass}}^{-1}. \end{aligned} \tag{6.8}$$

Let $J \in \mathbb{R}^{N\times 4}$ be the isometric block lift, with $J_{ia} = 1/\sqrt{m_a}$ for i in block a, and zero otherwise. Then

$$S = N J A_p J^*, \qquad G_p = \frac{N\lambda I_N - S}{N\lambda - 1} \tag{6.9}$$

is the required Gram matrix. In particular $S_{ij} = C_{ab}$ for $i \in a$, $j \in b$.

Proof. The Bloch identity gives $B_{aa} = \lambda - p_a$ and $|B_{ab}| = \sqrt{d_a d_b w_a w_b} = \sqrt{p_a p_b}$ for $a \neq b$. The matrix B is positive semidefinite of rank two. Indeed rank one would force all spinor lines, and hence all Bloch points, to coincide. Thus A_p has top eigenvalue λ of multiplicity exactly two, while $C_{aa} = 1$ and $|C_{ab}| = 1$. The spectrum of S is the spectrum of NA_p together with $N - 4$ zeros. Hence $N\lambda I - S$ is positive semidefinite with exactly two zero eigenvalues. Its diagonal is $N\lambda - 1$, and every off-diagonal entry has modulus one.

An explicit vector factor is obtained as follows. Choose $R \in \mathbb{C}^{N\times(N-4)}$ with orthonormal columns spanning $\mathrm{range}(J)^\perp$. Then

$$V_p = \frac{1}{\sqrt{N\lambda - 1}} \begin{pmatrix} \sqrt{N\lambda}\, R^* \\ \sqrt{N}\, W J^* \end{pmatrix} \in \mathbb{C}^{(N-2)\times N} \tag{6.10}$$

satisfies $V_p^* V_p = G_p$, because $RR^* = I - JJ^*$ and $N\lambda I - S = N\lambda(I - JJ^*) + NJBJ^*$. For $N = 4$ the first block of rows is empty. □

Remark 6.4 (Choices and equivalences)*.* The construction uses only the specified root, matrix square roots, and the displayed spinor lift. The first three Bloch vectors can be obtained by factoring their positive definite principal Gram matrix; the fourth is then determined by its inner products. Positivity of that principal matrix follows because the kernel of X has no vector with a zero coordinate except zero. Any two realizations of X differ by an orthogonal

transformation. A proper rotation corresponds to a common unitary on the spinors, and an improper rotation to a common unitary followed by conjugation. One checks the rotation assertion using $\cos(\theta/2)I - i\sin(\theta/2)n\cdot\sigma$; complex conjugation reverses the second Bloch coordinate. Independent spinor phases give diagonal unitary conjugations of B, C, S, G_p. Thus for an ordered profile G_p is unique up to diagonal unitary conjugation and possibly entrywise conjugation. Permuting the profile permutes the blocks.

Proposition 6.5 (A positive stress with strict phase alignment)**.** *For the above construction let*

$$\ell_a = \frac{2q_a}{d_a} = \frac{1-\lambda-p_a}{(1-\lambda)(\lambda-2p_a)}, \quad L = \operatorname{diag}(\ell_a), \quad T = L - LBL. \tag{6.11}$$

Then $T \succeq 0$ has rank two and range $\ker B$. Choose $U \in \mathbb{C}^{4\times 2}$ with $U^*U = I_2$ and $BU = 0$, and set

$$z_a = \frac{U^* e_a}{\sqrt{2p_a}}, \qquad Z = \sqrt{N/2}\,U^*J^*, \qquad K = \frac{2U^*TU}{\operatorname{tr} T}. \tag{6.12}$$

The matrix Z repeats z_a exactly m_a times and obeys $ZZ^* = NI_2/2$ and $ZS = N\lambda Z$. Moreover $K \succ 0$, $\operatorname{tr} K = 2$, and all entries of Z^*KZ are nonzero with entrywise phase S. Consequently

$$\Gamma_K(Z) := \sum_{i,j} |z_i^* K z_j| = N^2\lambda. \tag{6.13}$$

With K held fixed, Z is a critical point of

$$\mathcal{L}(Y) = \Gamma_K(Y) - N\lambda \operatorname{tr}(KYY^*). \tag{6.14}$$

Proof. Since $\sum q_a x_a = 0$ and $\sum q_a = 1$, the Bloch identity gives $\sum q_a u_a u_a^* = I_2/2$, and hence $WLW^* = I_2$. If $E = WL^{1/2}$, then $EE^* = I_2$, so E^*E is a rank-two orthogonal projector. Therefore

$$T = L^{1/2}(I - E^*E)L^{1/2} \succeq 0, \qquad \operatorname{rank} T = 2.$$

Also $WT = WL - (WLW^*)WL = 0$. Thus the range of T is $\ker B$, U^*TU is positive definite, and $\operatorname{tr}(U^*TU) = \operatorname{tr} T$. The exact entry formulas are

$$T_{ab} = \ell_a\ell_b\sqrt{p_ap_b}\,C_{ab} \quad (a \neq b), \qquad T_{aa} = \ell_a(1-2q_a) > 0.$$

All factors in these comparisons are strictly positive, by Lemma 6.2. As $UKU^* = 2T/\operatorname{tr} T$,

$$(Z^*KZ)_{ij} = \frac{T_{ab}}{\operatorname{tr} T\sqrt{p_ap_b}} \quad (i \in a,\ j \in b).$$

This proves the phase assertion even for distinct indices in the same block. The covariance and eigenvector identities follow from $J^*J = I$ and $BU = 0$. Thus

$$\Gamma_K(Z) = \operatorname{tr}(SZ^*KZ) = \operatorname{tr}(KZSZ^*) = N^2\lambda.$$

In a neighborhood of Z all weighted overlaps remain nonzero. If $C_K(Y)$ is their phase matrix, differentiation with the real Frobenius inner product gives $\nabla\Gamma_K(Y) = 2KYC_K(Y)$. At Z this is $2KZS = 2N\lambda KZ$, exactly the gradient of the constraint term in (6.14). □

Proposition 6.6 (Exact balancing)**.** *Among admissible integer four-weight profiles with total N, λ_p is largest and μ_p is smallest precisely when the four counts differ by at most one. For every $N \geq 10$ the balanced profile is admissible. If $N \equiv 2 \pmod 4$, then*

$$\lambda_{\mathrm{bal}} = \frac{1+\sqrt{3-8/N^2}}{4}, \qquad \mu_{\mathrm{bal}} = \frac{4}{N+\sqrt{3N^2-8}-4}. \tag{6.15}$$

Proof. For fixed $\lambda > 2/3$, the function $x \mapsto (\lambda - 2x)^{-1}$ is strictly convex. If two counts differ by at least two, transfer one from the larger to the smaller. All counts stay positive and below $N/3$, and the sum of reciprocal denominators strictly decreases. The coefficient $\lambda(3\lambda-2)$ in (6.2) is positive, so at the old root the new Φ is negative. Its unique root is strictly larger. Iteration terminates at balanced counts, proving also the equality characterization. For $N \geq 10$, $\lceil N/4 \rceil < N/3$. For residue two the weights are (a, a, b, b) with $a + b = 1/2$, $ab = 1/16 - 1/(4N^2)$, and

$$\sum_a \frac{1}{\lambda - 2p_a} = \frac{4\lambda - 2}{\lambda^2 - \lambda + 4ab}.$$

The root equation reduces to $2\lambda^2 - \lambda - 4ab = 0$; its relevant root is $(1 + \sqrt{1 + 32ab})/4$. □

6.1 The candidate domain needed at the endpoint

Write $p_a = 1/4 + \varepsilon_a$, $e = \max |\varepsilon_a|$ and $\sigma^2 = \sum \varepsilon_a^2$. Throughout this subsection $e \leq 1/120$. This includes every count profile in Proposition 4.2 for $N \geq 180$. All estimates concern the same alignment of the candidate and a reference tetrahedron.

The exact scalar identities are

$$\Phi_p(\alpha) = 4 \sum_a \frac{\varepsilon_a^2}{\alpha - 1/2 - 2\varepsilon_a}, \qquad \Phi_p'(t) = 4 \left[2 + \sum_a \frac{p_a(1 - 3p_a)}{(t - 2p_a)^2} \right].$$

For $f_t(p) = p(1-3p)/(t-2p)^2$,

$$f_t''(p) = \frac{2[t(4-3t) + 4p(1-3t)]}{(t-2p)^4} > 0$$

on $2/3 \leq t \leq \alpha < 41/60$, $29/120 \leq p \leq 31/120$: the numerator bracket is at least $43/200$. Jensen gives $\Phi_p'(t) > 4568/121 > 37$, whereas $4/(\alpha - 1/2 - 2e) < 1200/49 < 74/3$. Therefore

$$0 \leq \alpha - \lambda_p \leq \frac{2}{3}\sigma^2, \qquad \frac{109}{100} < b := 3(2\lambda_p - 1) < \frac{11}{10}, \qquad u_0 := \frac{3}{4} - \lambda_p > \frac{1}{15}. \tag{6.16}$$

For balanced counts, $N^2\sigma^2 = r(4-r)/4$. The odd budget is thus $1/2$. For residue two the exact formula gives $N^2(\alpha - \lambda_{\mathrm{bal}}) = 2/(\sqrt{3} + \sqrt{3 - 8/N^2}) < 11/19$ at $N \geq 180$. These are the budgets in (4.3).

6.2 Exact projector and simultaneous polar estimates

Let $\Pi = UU^*$ project onto $\ker B$. The two positive eigenvalues of B have sum $4\lambda - 1$ and product $D = 3\lambda(2\lambda - 1)$, whence

$$\Pi = I - \frac{(4\lambda - 1)B - B^2}{D}, \qquad \Pi_{aa} = \frac{3\lambda - 2 + 2p_a}{b}.$$

For complementary pairs $\{a, b\}, \{c, d\}$, completion of U to a unitary matrix and equality of complementary minors give

$$\det \Pi[\{a,b\}] = \det(I - \Pi)[\{c,d\}] = \frac{\det B[\{c,d\}]}{D} = \frac{\lambda - 1 + p_a + p_b}{b}.$$

These identities determine both the lengths $r_a = \|z_a\|$ and the Bloch Gram matrix Y_p of the normalized candidate spinors, relative to the tetrahedral Bloch Gram matrix $Y_0 = \frac{4}{3}I_4 - \frac{1}{3}\mathbf{1}\mathbf{1}^T$:

$$r_a^2 = \frac{1 + 4\varepsilon_a/b}{1 + 4\varepsilon_a}, \tag{6.17}$$

$$(Y_p - Y_0)_{ab} = \frac{-\frac{8}{3b}(\varepsilon_a + \varepsilon_b) + \frac{64}{3b^2}\varepsilon_a\varepsilon_b}{(1 + 4\varepsilon_a/b)(1 + 4\varepsilon_b/b)} \quad (a \neq b). \tag{6.18}$$

The diagonal of $Y_p - Y_0$ is zero. The zero-sum identities imply

$$G := \|Y_p - Y_0\|_F \leq \frac{16/(3b) + 112e/(3b^2)}{(1 - 4e/b)^2}\,\sigma \leq \frac{11}{2}\sigma \leq \frac{11}{120}.$$

The coefficient at $b = 109/100, e = 1/120$ is $551200/100489 < 11/2$.

Here is a polar comparison used also for the actual group means. For a real 3×4 unit-column matrix X and a tetrahedron V with $V\mathbf{1} = 0$, $VV^T = 4I/3$, set $C_0 = 3XV^T/4$, $c = X\mathbf{1}/4$. Then $X = C_0V + c\mathbf{1}^T$ and, if $G = \|X^TX - V^TV\|_F$,

$$G^2 = \frac{16}{9}\|C_0^TC_0 - I\|_F^2 + \frac{32}{3}\|C_0^Tc\|^2 + 16\|c\|^4. \tag{6.19}$$

This follows by expanding and using $V\mathbf{1} = 0$. Also $s_{\min}(C_0)^2 \geq 1 - 3G/4$. For the orthogonal polar factor R,

$$\|X - RV\|_F^2 \leq P_G G^2, \qquad P_G = \max\left\{\frac{3/4}{(1 + \sqrt{1 - 3G/4})^2}, \frac{3/8}{1 - 3G/4}\right\}. \tag{6.20}$$

Indeed compare the two nonnegative terms $\frac{4}{3}\|C_0 - R\|_F^2 + 4\|c\|^2$ with the first two terms of (6.19). At $G \leq 11/120$, $s_{\min}(C_0)^2 \geq 149/160$, so $P_G < 41/100$. An orientation reversal is handled by conjugating the reference spinors.

For unit spinors v, f and their Bloch vectors x, y,

$$d_{\mathrm{ph}}(v, f)^2 = 2(1 - |v^*f|) = \frac{\|x - y\|^2}{2(1 + |v^*f|)}. \tag{6.21}$$

The total bound $(41/100)(11/120)^2 < 4[1 - (999/1000)^2]$ allows one common polar alignment and phases with $|v_a^*f_a| > 999/1000$. Moreover

$$|r_a^2 - 1| \leq \frac{4e}{11(1 - 4e)} \leq \frac{1}{319}, \qquad \frac{499}{500} < r_a < \frac{501}{500},$$
$$|r_a - 1| \leq c_r|\varepsilon_a|, \qquad c_r = \frac{4/11}{(1 - 4e)(1 + 499/500)}.$$

All these assertions use the larger $1/120$ domain; no smaller-domain polar floor is extrapolated.

For the actual four count profiles, the following explicit formulas sharpen the simultaneous errors. Put $\varepsilon_a = \delta_a/N$, $\sigma = \sqrt{C}/N$, $b_0 = 109/100$, $k_0 = 8/(3b_0)$ and $q_0 = 1 - 4e/b_0$. Write $\Delta_{\mathrm{B}} = Y_p - Y_0 = D_0 + F$, where $(D_0)_{ab} = -8(\varepsilon_a + \varepsilon_b)/(3b)$ off the diagonal. Exact subtraction in (6.18) gives

$$F_{ab} = \frac{32}{3b^2}\,\frac{\varepsilon_a^2 + 4\varepsilon_a\varepsilon_b + \varepsilon_b^2 + (4/b)(\varepsilon_a + \varepsilon_b)\varepsilon_a\varepsilon_b}{(1 + 4\varepsilon_a/b)(1 + 4\varepsilon_b/b)}.$$

Thus $|F_{ab}| \leq U_{ab}$, where

$$U_{ab} = \frac{32}{3b_0^2N^2q_0^2}\left(|\delta_a^2 + 4\delta_a\delta_b + \delta_b^2| + \frac{4}{b_0N}|(\delta_a + \delta_b)\delta_a\delta_b|\right).$$

Let $f = (\sum_{a\neq b} U_{ab}^2)^{1/2}/\sigma$. To see explicitly how this supplies a common polar bound, put $E = 9V\Delta_{\mathrm{B}}V^T/16$, $z = 3V\Delta_{\mathrm{B}}\mathbf{1}/16$ and $P_0 = (I + E)^{1/2}$. The aligned Bloch error is $H = (P_0 - I)V + P_0^{-1}z\mathbf{1}^T$. Its linear parts satisfy

$$E_0 = \frac{9k}{8}V\,\mathrm{diag}(\varepsilon_a)V^T, \quad z_0 = -\frac{3k}{8}V\varepsilon, \quad k = 8/(3b),$$

$$\|E_0\|_F = \frac{3\sqrt{2}k}{4}\sigma, \quad |z_0| = \frac{\sqrt{3}k}{4}\sigma.$$

These follow from $V^TV = \frac{4}{3}I - \frac{1}{3}\mathbf{1}\mathbf{1}^T$ and $\sum \varepsilon_a = 0$. Set

$$K_E = \frac{3\sqrt{2}}{4}k_0 + \frac{3}{4}f, \quad K_z = \frac{\sqrt{3}}{4}(k_0 + f), \quad p_0 = \sqrt{1 - K_E\sigma},$$

$$R_P = \frac{K_E^2\sigma}{2(1+p_0)^2}, \qquad R_I = \frac{K_E K_z \sigma}{p_0(1+p_0)}.$$

The scalar square-root identity and $P_0^{-1} - I$ give these full nonlinear remainders. The tetrahedral Gram identities, applied separately to each column, then give

$$|H_a|/\sigma \le \frac{\sqrt{3 - 4\delta_a^2/C}}{b_0} + \frac{13}{16}f + R_P + R_I,$$
$$\|H\|_F/\sigma \le \frac{\sqrt{8}}{b_0} + \frac{\sqrt{15}}{4}f + \frac{2}{\sqrt{3}}R_P + 2R_I.$$

For clarity, the linear point norm equals $\sqrt{3\sigma^2 - 4\varepsilon_a^2}/b$; its aggregate norm is $\sqrt{8}\,\sigma/b$. The nonlinear F contribution is bounded by $13\|F\|_F/16$ per column and $\sqrt{15}\|F\|_F/4$ in aggregate, using $E_F = 9VFV^T/16$, $z_F = 3VF\mathbf{1}/16$ and $\|E_F\|_F \le 3\|F\|_F/4$, $|z_F| \le \sqrt{3}\|F\|_F/4$. The two arrays $E_F V/2$ and $z_F\mathbf{1}^T$ are orthogonal, giving the aggregate squared coefficient $15/16$; the point coefficient is $3/8 + \sqrt{3}/4 < 13/16$. This explains the paired, rather than independently maximized, column dependence.

With $\phi = (501/500)/[2(1 + 999/1000)]$ and $R_* = (13/16)f + R_P + R_I$, define

$$C_{\mathrm{pt},a} = c_r^2\delta_a^2/C + \phi[\sqrt{3 - 4\delta_a^2/C}/b_0 + R_*]^2,$$
$$C_{\mathrm{pt}} = \max_a C_{\mathrm{pt},a},$$
$$B_p = (1/4 + e)\left\{c_r^2 + \phi[\sqrt{8}/b_0 + (\sqrt{15}/4)f + (2/\sqrt{3})R_P + 2R_I]^2\right\}.$$

In this same alignment they prove

$$\max_a \|z_a - f_a\|^2 \le C_{\mathrm{pt}}C/N^2, \qquad \sum_a p_a\|z_a - f_a\|^2 \le B_pC/N^2. \tag{6.22}$$

All divisions are guarded by $p_0 > 0$. On the full domain the generic bound $f < 61/100$ implies $K_E < 33/8$, $K_z < 77/32$, hence positivity; actual profile bounds are sharper. Both signs and every permutation are included in the absolute polynomials U_{ab}.

Finally the stress norm has a basis-free estimate. Let $j_a = 1/\ell_a$ and $H = U^*\,\mathrm{diag}(j_a)U$. The identity $TL^{-1}\Pi = \Pi$ gives $U^*TU = H^{-1}$ and $\|K - I\| = \mathrm{gap}(H)/\operatorname{tr} H$. Writing $j_c = j(1/4, \lambda)$, exact division gives

$$j_a/j_c = 1 + C_0\varepsilon_a + C_0\varepsilon_a^2/(u_0 - \varepsilon_a), \quad C_0 = \frac{3\lambda - 2}{(\lambda - 1/2)u_0}, \quad 0 \le C_0 \le 4.$$

For $E = U^*\,\mathrm{diag}(\varepsilon_a)U$, the projector minors give

$$\operatorname{tr} E = 2\sigma^2/b, \qquad \mathrm{gap}(E)^2 = \sigma^2/3 + 4\sum \varepsilon_a^3/b + 4\sigma^4/b^2.$$

The remaining compression is positive semidefinite and has spectral gap at most $4e^2/(u_0 - e)$, whereas $\operatorname{tr}(H/j_c) \ge 2$. Consequently

$$\kappa := \|K - I\| \le 2\sigma\sqrt{1/3 + 4e/b_0 + 4\sigma^2/b_0^2} + \frac{2e^2}{1/15 - e} < \frac{11}{4}e. \tag{6.23}$$

The last endpoint check is $4\sqrt{39013/106929} + 2/7 < 11/4$. The radial and spinor estimates also give a generic bound η for the candidate point error $\max_a \|z_a - f_a\|$: $\eta^2 \le [500/(319 \cdot 999)]^2 + \phi(41/100)(11/120)^2 < 1/1024$. The profile-dependent choice of η is specified in (8.3). Thus the candidate point error and κ are below $1/32$. For a fixed count profile all analytic upper coefficients above improve as N increases. Freezing valid endpoint enclosures therefore proves the whole subsequent range; no claim about monotonicity of rounded program outputs is needed.

7 Localization of every column at the endpoint 180

This section proves the global and fine inputs to Proposition 4.2. All partitions below contain every original column. The same partition is kept during the forward estimates.

7.1 Deficit identity and estimates before partitioning

Let $t_{ij} = (1 + x_i \cdot x_j)/2$. The polynomial

$$f(t) = \alpha + \tfrac{3}{4}(t - \tfrac{1}{2}) - k(t^2 - t + \tfrac{1}{6})$$

satisfies $f(t) - \sqrt{t} = g(t)$ exactly. Put $T_1 = \sum_{ij} r_i r_j t_{ij}$. Tightness gives $\sum_i r_i^2 t_{ij} = N/2$, so

$$N^2/2 - T_1 = \tfrac{1}{2}\sum_{ij}(r_i - r_j)^2 t_{ij} \ge 0.$$

Also $\sum_{ij} r_i r_j (t_{ij}^2 - t_{ij} + 1/6) = \frac{1}{4}\|\sum r_i x_i x_i^T - (L/3)I\|_F^2$. Summing the polynomial identity proves

$$\alpha N^2 - \Gamma_I(Y) = u + v + w + \tfrac{3}{4}(N^2/2 - T_1). \tag{7.1}$$

Every term on the right is nonnegative. In particular $L \le N$.

Set $c = 13/4$. On $N \ge 180$, $u + v + w \le M \le 11/19$ gives $L/N > 9999/10000$, using $a > 77/250$. Since $(77/250)(19999/10000) > 8/13$,

$$D_r := \sum (r_i - 1)^2 = \frac{2u}{a(N+L)} \le \frac{cu}{N}, \quad s = \sqrt{cu/N} < \frac{103}{1000}. \tag{7.2}$$

Thus every radius is positive. With $E = \sum_{ij} g(t_{ij})$,

$$E \le \frac{w}{(1-s)^2} \le \frac{M}{1 - cM/N} < \frac{293}{500}. \tag{7.3}$$

The second bound retains $u + w \le M$: after setting $x = \sqrt{cu/N}$, $m = cM/N$, its numerator difference is $m(1-x)^2 - (m - x^2)(1 - m) = (x - m)^2$.

Put $R_x = \sum x_i$, $M_0 = \sum x_i x_i^T$, and $l_i = (1, \sqrt{3}x_i)^T/2 \in \mathbb{R}^4$. Cauchy–Schwarz and $\sum r_i^2 x_i = 0$ give

$$\|M_0 - NI/3\|_F \le 2\sqrt{v/k} + \sqrt{2cu/3}, \quad \|M_0 - NI/3\|_{\mathrm{op}} < 23/10,$$

$$\|R_x\|^2 \le (N/3 + 23/10)(2 + s)^2 D_r.$$

For the operator estimate, a traceless symmetric 3×3 matrix has squared operator norm at most two thirds of its squared Frobenius norm, and $(2/3)(1000/87 + 13/6)M < (23/10)^2$. The exact lifted identity is

$$\Big\|\sum l_i l_i^T - NI_4/4\Big\|_F^2 = \tfrac{3}{8}\|R_x\|^2 + \tfrac{9}{16}\|M_0 - NI_3/3\|_F^2 \le T(u, v) := Au + Bv + J\sqrt{uv},$$

where

$$A = \frac{19739570097}{6400000000}, \quad B = \frac{375}{58}, \quad J = \frac{1123}{200}, \quad T(u,v) \le \frac{81}{10}(u+v). \tag{7.4}$$

The last quadratic comparison has positive diagonal entries and determinant

$$\frac{293253687011}{928000000000} > 0.$$

If $z = u + v$, the traceless 4×4 spectral inequality therefore yields

$$\lambda_{\min}\Big(\sum l_i l_i^T\Big) \ge \Lambda_N(z) := N/4 - \sqrt{243z/40} > 0. \tag{7.5}$$

Indeed an eigenvalue t of a traceless 4×4 real symmetric matrix forces the other squared eigenvalues to sum to at least $t^2/3$.

7.2 Simultaneous anchors and coverage

Let $\mathcal{L} = [l_1 \ \cdots \ l_N]$ and $F = \mathcal{L}\mathcal{L}^T$. Give each four-element subset A probability $\det(\mathcal{L}_A)^2 / \det F$. For a set J of size one or two,

$$\Pr(J \subset A) = \det(\mathcal{L}_J^T F^{-1} \mathcal{L}_J). \tag{7.6}$$

To verify this identity, whiten $U = F^{-1/2}\mathcal{L}$, so $UU^T = I$. Cauchy–Binet expands $\det(U \operatorname{diag}(t_i) U^T) = \sum_{|A|=4} \det(U_A)^2 \prod_{i \in A} t_i$. Taking the derivatives indexed by J at $t_i = 1$, and alternatively expanding $\det(I + U \operatorname{diag}(t_i - 1) U^T)$, gives (7.6).

Write $q(t) = (3t-1)/2$, so $l_i^T l_j = q(t_{ij})$. Loewner order in the one- and two-point determinants gives

$$\Pr(i \in A) \le \Lambda_N(z)^{-1}, \qquad \Pr(i,j \in A) \le [1 - q(t_{ij})^2] \Lambda_N(z)^{-2}.$$

The scalar bounds $q(t)^2 \le 24 g(t)$ for $t < 2/3$ and $1 - q(t)^2 \le 96 g(t)$ for $t \ge 2/3$ follow from the canceled ratios below, their derivatives and their values at $2/3$. Explicitly, $g(t)/q(t)^2$ is at least its value at $2/3$, which exceeds $1/24$, on the lower interval. On the upper interval $g(t)/(1-t) > 1/32$ and $1 - q(t)^2 = (3/4)(1-t)(1+3t) \le 3(1-t)$. Endpoints with zero factors are handled directly or by continuity. If E_A is the sum of selected angular row energies, H counts selected unordered pairs with $t \ge 2/3$, and $X = 2 \sum_{\{i,j\} \subset A, t_{ij} < 2/3} q(t_{ij})^2$, then

$$\mathbb{E} E_A \le E/\Lambda_N(z), \quad \mathbb{E} X \le 24E/\Lambda_N(z)^2, \quad \mathbb{E} H \le 48E/\Lambda_N(z)^2.$$

Let $K_A = 441/100$, $\eta_A = 8/25$. For $E > 0$, the expected potential $E_A/(K_A E/N) + X/\eta_A^2 + H$ is at most

$$\frac{N}{K_A \Lambda_N(z)} + \Big(24/\eta_A^2 + 48\Big) \frac{(M-z)/(1-\sqrt{cz/N})^2}{\Lambda_N(z)^2} < 1. \tag{7.7}$$

The strict inequality is certified on 256 closed intervals covering $0 \le z \le M$ at $N = 180$, as specified in Section 9. For each interval $[l,h]$, replace Λ by $45 - \sqrt{243h/40}$ and the energy by

$$\min\{M/(1 - cM/180), (M-l)/(1 - \sqrt{ch/180})^2\}.$$

Every denominator is positive. This is a closed interval proof, not a sampled maximization. Thus one actual four-subset has $H = 0$, $x = \sqrt{X} < \eta_A$ and $E_A < (K_A E/N)(1 - x^2/\eta_A^2)$.

The selected lifted Gram has smallest eigenvalue at least $1 - \sqrt{3}x/2$. The exact identity

$$\frac{2}{1+\sqrt{1-h}}(1-y) - (1 - y^2/h) = \frac{(y - 1 + \sqrt{1-h})^2}{h} \ge 0$$

gives the coverage and disjointness ratios

$$R_{\mathrm{cov}} = \frac{2}{1+\sqrt{1-3\eta_A^2/4}}, \qquad R_{\mathrm{dis}} = \frac{2}{1+\sqrt{1-\eta_A^2/2}}.$$

For $h = \sqrt{t}$ and $b = 1/\sqrt{3}$,

$$\frac{g(t)}{q(t)^2} = \frac{4(1-h)(3/4+kh)}{9(h+b)^2}, \qquad \frac{g(t)}{1-t} = \frac{(h-b)^2(3/4+kh)}{1+h}.$$

The first ratio decreases, with derivative $-4\sqrt{3}/[9(h+b)^3]$; the second increases for $h \geq b$. The exact endpoint checks are

$$R_{\mathrm{cov}}\frac{K_A(293/500)}{180} < \frac{g(173/200)}{q(173/200)^2}, \qquad R_{\mathrm{dis}}\frac{K_A(293/500)}{180} < \frac{1}{50}.$$

If a data column were outside every 173/200 cap, its unit lifted vector would give $\sum_a q(t_{iA_a})^2 \geq 1-\sqrt{3}x/2$; its incident energies would then contradict the first check. Two 83/100 memberships of an actual data column would imply an anchor squared distance at most $(400/3)E_A$, using $g(t)/(1-t) > 3/50$ there, whereas the selected-pair bound gives at least $(8/3)(1-x/\sqrt{2})$. The second check excludes it. These assertions concern the data columns; the continuous caps need not be disjoint. Since $173/200 > 83/100$, they give one partition into four nonempty groups.

Let $\bar{x}_a$ be the means and $S = \sum_a \sum_{i\in a} \|x_i - \bar{x}_a\|^2$. A valid lower membership ratio at 173/200 is

$$m_0 = \frac{667855014484621414278044848317828318861}{9650268809435000000000000000000000000000}.$$

Since means minimize squared distance, $S \leq 4E_A/m_0 < 4K_AE/(m_0N) < 255E/N$. We may weaken this to $S \leq 264E/N$ for the subsequent common bootstrap. If $E = 0$, all t_{ij} belong to $\{1/3, 1\}$. The positive lifted frame bound supplies four spanning data columns; their lifted vectors are orthonormal. Every other unit lifted vector has inner products in $\{0, 1\}$ with that basis, and is one of them. This gives the same partition and $S = 0$ without division by E.

7.3 Counts before fine localization

Let Σ_a be the within-group covariance, $S_a = \operatorname{tr}\Sigma_a$, $\delta_a = m_a - N/4$, and let the square matrix A_0 have columns $\sqrt{m_a}(1, \sqrt{3}\bar{x}_a)^T/2$. If $F_0 = \sum l_i l_i^T - NI/4$, $H_0 = \operatorname{diag}(0, \sum \Sigma_a)$ and $t_0 = (N - 3S/4)/4$, then

$$A_0A_0^T - t_0I = F_0 - \tfrac{3}{4}(H_0 - SI/4), \quad \operatorname{diag}(A_0^TA_0 - t_0I) = \delta - \tfrac{3}{4}(S_a - S/4)_a.$$

The two Gram matrices have identical spectra. Their Frobenius norms, diagonal contraction, and the centered PSD estimates imply

$$\sqrt{C} \leq \sqrt{T(u,v)} + \frac{3\sqrt{3}}{4}S \leq \sqrt{T(u,v)} + \frac{13}{10}S. \tag{7.8}$$

Each centered norm is at most $\sqrt{3}S/2$; both variance terms are necessary. With $S \leq 264E/N$ and the shared budget, 256 closed z-interval checks give $C < 5$ for $M = 11/19$ and $C < 19/4$ for $M = 1/2$. This precedes, and therefore does not rely on, any fine estimate.

For residue two the δ_a are half-integers with zero sum. The strict square bound allows only $|\delta_a| \leq 3/2$, with at most one entry of that magnitude, giving the two listed profiles. For odd N, after sign reversal let $4\delta_a \equiv -1 \pmod 4$. Their squared sum is below 76. Magnitudes at least 9 are excluded; an entry 7 requires other entries summing to -7, whose minimum squares are $25 + 1 + 1$, giving total 76. Thus only $-5, -1, 3$ remain. Their zero sum permits either no -5, giving $(-1, -1, -1, 3)$, or one -5, giving $(-5, -1, 3, 3)$. Two -5 require two entries 3 summing only to 6, so cannot have zero sum. This proves the complete count list.

7.4 Shared-budget forward refinement

Assume $S \le c_s E/N$ has already been proved on this partition. For a closed region $u \le U, w \le W, u+v+w \le M$, let $\bar{E}$ be a valid energy cap and set

$$p = 1/4 - m_c/N, \quad p_+ = 1/4 + m_c/N, \quad S_* = c_s\bar{E}/N, \quad q = \sqrt{1 - S_*/(pN)}, \quad T_* = 2S_*/(1+q),$$

where $m_c = \max|\delta_a|$. All square roots and divisions require their displayed positive domains. A sharp cap $\bar{E}$ follows by maximizing $\min(W, M-u)/(1-\sqrt{cu/N})^2$ over $0 \le u \le U$. On the branch $w = M-u$ the sole interior maximum is $u = (c/N)M^2$, of value $M/(1-cM/N)$; otherwise use the boundary.

Write $Q = \sum_{a \ne b} m_a m_b (\bar{x}_a \cdot \bar{x}_b + 1/3)^2$. The off-diagonal part of the preceding square mean-Gram calculation gives $\sqrt{Q} \le (4/3)\sqrt{T} + S$. Write $c_\times = \max_{a(i) \ne a(j)}(x_i \cdot x_j + 1/3)$. Normalizing the means gives the following three bounds, kept under a single budget supremum:

$$\begin{aligned} c_\times &\le \frac{4\sqrt{T}}{3Np} + b_s\sqrt{E} + d_s E, \\ G &\le \frac{4\sqrt{T}}{3Np} + d_s E, \qquad \frac{\sqrt{Q}}{pN} \le \frac{4\sqrt{T}}{3Np} + d_m E, \\ b_s &= \sqrt{\frac{4c_s}{N(1+q)}}, \quad d_m = \frac{c_s}{N^2 p}, \quad d_s = d_m\left(1 + \frac{\sqrt{6}}{1+q}\right). \end{aligned} \tag{7.9}$$

Here G bounds the Frobenius Gram error of the four normalized means. The normalization costs follow from $\sum_i \|x_i - \bar{x}_a/\|\bar{x}_a\|\|^2 \le 2S/(1+q)$.

On a closed bin $z \in [z_l, z_h]$, use $u \le \min(U, z_h)$ and $w \le \min(W, M - z_l)$ to bound the energy and the *whole* expressions (7.9). The raw moment $Au + Bv + J\sqrt{uv}$ can be maximized on $u + v = z_h$ by its concave one-variable expression. Its unconstrained upper bound is $z_h(A + B + \sqrt{(A-B)^2 + J^2})/2$; if $U_0 = \min(U, z_h)$ obeys

$$2U_0 \le z_h, \quad J^2(z_h - 2U_0)^2 \ge 4(B-A)^2 U_0(z_h - U_0),$$

the constrained maximum is the value at U_0. This exact squared derivative test avoids an uncertain radical branch.

There is also a spherical improvement to the cross bound. If the unit mean dot product is at most $\gamma = -1/3 + G$ and the two squared chord errors sum to $T_s < 2$, concavity of $\arccos(1 - t/2)$ gives an angle error at most $4\arcsin\sqrt{T_s/8}$. Put

$$C_s = 1 - T_s + T_s^2/8, \quad S_s = \sqrt{2T_s(1 - T_s/8)}(1 - T_s/4).$$

When $\gamma < C_s$, the pair dot product is at most $\gamma C_s + \sqrt{1-\gamma^2}S_s$; otherwise use 1. Here $T_s = 2c_s E/[N(1+q)]$. This retains all within-group residuals. On each closed budget bin, enclose both the linear bound for $c_\times$ in (7.9) and the spherical bound just obtained, adding $1/3$ to the latter dot-product bound. If the angular conditions fail, use the trivial spherical bound $4/3$. Take the smaller of the two upper enclosures on each bin, with outward rational rounding, then the maximum over all bins covering the region. This gives a regional upper bound. It may be reduced by taking its minimum with other independently valid upper bounds on the same region. Denote the resulting bound by $\bar{c}_\times$, so $c_\times \le \bar{c}_\times$ throughout the region.

Set $h_W = \sqrt{1 - S_*/2}$ and $h_C = \sqrt{(2/3 + \bar{c}_\times)/2}$. The canceled scalar kernels are

$$K_W = \frac{(h_W - 1/\sqrt{3})^2(3/4 + kh_W)}{4(1+h_W)}, \quad K_C = \frac{(1-h_C)(3/4 + kh_C)}{4(h_C + 1/\sqrt{3})^2}.$$

Use lower enclosures of these kernels, $h_W > 1/\sqrt{3}$, $h_C < 1$, and

$$\mu_c = 2\left[\frac{1}{3} - \frac{23/10 + S_*}{N} - \frac{p_+ T_*}{2(1+q)}\right], \quad \beta = 2pK_W + \mu_c K_C > 0. \tag{7.10}$$

Then

$$E \geq K_C Q + \beta NS. \tag{7.11}$$

For detail, the cross-pair squared-dot sum is exactly

$$Q + 2\sum_a \operatorname{tr}\left[\left(\sum_{b\neq a} m_b \bar{x}_b \bar{x}_b^T\right)\Sigma_a\right] + \sum_{a\neq b}\operatorname{tr}(\Sigma_a\Sigma_b).$$

The last term is nonnegative. The full covariance is at least $(N/3 - 23/10 - S)I$. The variance of $x_i \cdot (\bar{x}_a/\|\bar{x}_a\|) \in [1 - T_*/2, 1]$ gives its removed same-group term at most $T_* S_a/[2(1+q)]$. Indeed if its mean is $r_a = \|\bar{x}_a\|$, the scalar variance is at most $(T_*/2)(1-r_a)$ and $m_a(1-r_a) = S_a/(1+r_a) \leq S_a/(1+q)$. These facts yield $\mu_c NS$ for the cross part. The within-group identity $\sum_{i,j\in a}\|x_i - x_j\|^2 = 2m_a S_a$ supplies $2pK_W NS$, proving (7.11).

Starting from 264, the recorded closed-domain inequalities prove

$$264 \longrightarrow 240 \longrightarrow 183 \longrightarrow 107 \longrightarrow 58 \longrightarrow 40 \longrightarrow 34 \longrightarrow 32 \longrightarrow 31. \tag{7.12}$$

Every arrow checks $\beta > 1/c_{\text{next}}$, hence is a forward implication, not an assumed fixed point. The uniform chain uses $m_c = 3/2$ and 128 closed budget bins per stage. Subsequent regional refinement starts from 31, uses the already established profile's m_c consistently, and uses 64 closed bins. No new partition is selected.

7.5 Fine polar and spinor interface

At the final regional coefficient let $h_m = \sqrt{(2/3 + \sqrt{Q}/(pN))/2}$, using its certified upper cap, and put $K_m = K_C(h_m)$. If $\gamma \leq 1/(4h_C^3) - 2k$ is positive, then $g(t) - \gamma t^2/2$ is convex on $[0, h_C^2]$. Indeed $g''(t) = 1/(4t^{3/2}) - 2k \geq \gamma$ for $t > 0$, and continuity extends convexity to 0. Equivalently, g is γ-strongly convex on this interval. Jensen's inequality gives

$$E \geq K_m Q + [2pK_W + \mu_c\gamma/8]NS.$$

Take a convex combination with (7.11), obtaining $E \geq c_Q Q + c_S NS$. If the positivity guard fails, use the first inequality alone. The factor $1/8$ is the product of the strong-convexity factor $1/2$ and the change of variable $t = (1 + x\cdot y)/2$ squared.

Use the single polar alignment of the normalized means, with P_G from (6.20) and $G < 1/12$. The mean/residual decomposition and Young's inequality with $\tau = 201/200$ give

$$\sum_i \|x_i - v_{a(i)}\|^2 \leq A_S S + A_Q Q/N,$$

$$A_Q = p_+ P_G \tau/p^2, \qquad A_S = \frac{2}{1+q} + \frac{6p_+ P_G(1 + 1/(\tau-1))S_*}{p^2 N(1+q)^2}.$$

Therefore $L_B = \max(A_Q/c_Q, A_S/c_S)$ proves a total Bloch error at most $L_B E/N$. This uses one joint covariance inequality, rather than independently maximizing Q and S.

Choose a phase floor f from $127/128, 63/64, 31/32$ only if all three strict guards hold:

$$L_B\bar{E}/N < 4(1-f^2), \quad T_* < 4(1-f^2), \quad P_G G^2 < 4(1-f^2). \tag{7.13}$$

They concern, respectively, actual-to-polar, actual-to-normalized-mean and normalized-mean-to-polar spinors. With $D_f = 1/[2(1+f)]$, the exact identity (6.21) and the quotient triangle inequality give

$$L_U = D_f L_B, \tag{7.14}$$
$$L_P = \left[\sqrt{\frac{2D_f}{(1+q)\beta} + \frac{D_f P_G}{p^2 N K_C}} + \frac{\sqrt{D_f P_G}\sqrt{6}\sqrt{\bar{E}/N^3}}{p(1+q)\beta}\right]^2 .$$

For the point estimate, the actual-to-mean contribution is bounded by $2S/(1+q)$; the mean-to-polar contribution uses $G \le \sqrt{Q}/(pN) + \sqrt{6}S/[pN(1+q)]$. Maximizing their joint square-root terms under $K_C Q + \beta N S \le E$ gives exactly the first square root in (7.14); the remaining normalization term gives the second. Thus no separate maximum of S and Q is used in the first term.

In one common preliminary alignment and choice of column phases,

$$N\|Y - F\|_F^2 \le A_R := cU + L_U W \frac{1+s_U}{(1-s_U)^2}, \tag{7.15}$$
$$N \max_i \|y_i - f_{a(i)}\|^2 \le P_R := cU + L_P W \frac{1+s_U}{(1-s_U)^2}, \qquad s_U = \sqrt{cU/N}.$$

Indeed $d_{\mathrm{ph}}(r_i u_i, f)^2 = (r_i - 1)^2 + r_i d_{\mathrm{ph}}(u_i, f)^2$. The sharper angular inputs are $\sum d_{\mathrm{ph}}(u_i, f)^2 \le L_U \bar{E}/N$ and $\max d_{\mathrm{ph}}(u_i, f)^2 \le L_P \bar{E}/N$. The actual closest-gauge transport, required for the local Hessian, is proved in Section 8; a convenient preliminary alignment is not substituted for that gauge.

7.6 Continuous range and zero budgets

Every accepted region is closed and all its constants cover its whole interior and boundary. For larger N, at fixed deficits the energy cap decreases, $N/\Lambda_N(z)$ decreases, and $\Lambda_N(z)$ increases. All anchor and count inequalities therefore propagate. In the fine chain, freeze the endpoint count floors, coefficient choices, closed bins and phase floors. The energy and scatter decrease, mean lengths improve, and every positive N^{-1} or $N^{-1/2}$ error decreases. The spherical cap is monotone geometrically because decreasing the allowed mean dot product or chord error shrinks the feasible angle range. Its algebraic maximizing branch need not remain fixed. Propagate the finite chain in order, then its regional refinement and polar interface. If $E = 0$, (7.11) gives $S = Q = 0$; all zero-error cases are direct. There is no division by u, v, w or E in the fine or local interface. These facts justify freezing the saved 180 localization coefficients for the later 182 local comparison.

8 Local comparison

This section proves the local implication used after Sections 6 and 7. All four profiles use the intrinsic residual and physical normal estimates. For the odd profiles the complete unsigned comparison suffices. For $C = 3$ we use the stress-coordinate mean estimate with the complete unsigned remainder. For $C = 1$ we use the refined row coefficients and the minimum of the complete signed alternatives and the complete unsigned bound. All derivatives are real derivatives on complex vector spaces, and $\langle H, J\rangle = \mathrm{Re}\,\mathrm{tr}(H^* J)$. Matrix norms without subscripts are operator norms. Frame distances are measured in the Frobenius metric on the original variables.

8.1 Inputs, gauge, and the normal decomposition

Let $a(i) \in \{1,2,3,4\}$ be the partition supplied by global localization, let $m_a = |\{i : a(i) = a\}|$, and put $p_a = m_a/N$. The candidate $Z = (z_{a(i)})_i$ and its positive stress K satisfy

$$ZZ^* = \frac{N}{2} I_2, \quad K > 0, \quad \operatorname{tr} K = 2, \quad S_{ij} = \frac{z_i^* K z_j}{|z_i^* K z_j|}, \quad ZS = N\lambda Z. \tag{8.1}$$

Every stress overlap is nonzero. Put $\alpha = (1+\sqrt{3})/4$. In one common alignment with unit tetrahedral spinors f_a, the candidate estimates take the form

$$e = \max_a |p_a - \tfrac{1}{4}|, \qquad \sigma^2 = \sum_a (p_a - \tfrac{1}{4})^2 = C/N^2, \qquad \|K - I\| \le \kappa, \qquad 0 \le \alpha - \lambda \le \tau, \tag{8.2}$$

$$\max_a \|z_a - f_a\| \le \eta, \qquad \sum_a p_a \|z_a - f_a\|^2 \le B\sigma^2, \qquad \eta = \sqrt{C_{\mathrm{pt}} C}/N. \tag{8.3}$$

Here B, C_{pt} are the simultaneous polar bounds of Section 6; the separate point and aggregate estimates must use the same alignment. The four surviving profiles are

C	$3/4$	$11/4$	1	3
$m_* = \max_a \lvert m_a - N/4\rvert$	$3/4$	$5/4$	$1/2$	$3/2$
$L_* = \sum_a \lvert m_a - N/4\rvert$	$3/2$	3	2	3

with both signed profiles and all permutations included. Throughout the certified range, $e \le 1/120$ and $\eta, \kappa < 1/32$. Write $p_- = 1/4 - e$, $p_+ = 1/4 + e$, and $\omega_p = 1 - 4e > 0$.

The competitor is any normalized left kernel Y with $YY^* = NI_2/2$ and $YA = 0$. There is no assumption that its columns have equal lengths or take only four values. On a closed budget region, global localization supplies nonnegative numbers $A_R, P_R, L_U, L_P, \bar{E}$ such that, in its preliminary alignment,

$$\|Y - F\|_F^2 \le A_R/N, \qquad \max_i d_{\mathrm{ph}}(y_i, f_{a(i)})^2 \le P_R/N, \tag{8.4}$$

$$\sum_i d_{\mathrm{ph}}(u_i, f_{a(i)})^2 \le L_U \bar{E}/N, \qquad \max_i d_{\mathrm{ph}}(u_i, f_{a(i)})^2 \le L_P \bar{E}/N, \tag{8.5}$$

where $u_i = y_i/\|y_i\|$ and d_{ph} minimizes over a scalar phase. The first norm is evaluated in the preliminary phase choices. The physical lengths lie in $[1-s, 1+s]$, with $s < 1$. These bounds hold throughout the region.

For fixed K define

$$\Gamma_K(Y) = \sum_{i,j} |y_i^* K y_j|, \qquad \mathcal{L}(Y) = \Gamma_K(Y) - N\lambda \operatorname{tr}(KYY^*), \qquad Q(H,J) = \tfrac{1}{2} D^2 \mathcal{L}(Z)[H,J].$$

When one argument is displayed, $Q(H)$ means $Q(H,H)$.

Lemma 8.1 (Closest gauge)**.** *Choose a minimizer of $\|UYD - Z\|_F$ over $U \in U(2)$ and diagonal unitary D, and relabel that representative as Y. Set $\Delta = Y - Z$, $d = \|\Delta\|_F$, and*

$$C_n = \Delta\Delta^*/N, \qquad R_n = C_n Z, \qquad \Xi = \Delta + R_n.$$

Then

$$D\mathcal{L}(Z) = 0, \qquad \Gamma_K(Z) = N^2\lambda, \qquad \Delta Z^* = Z\Delta^* = -\tfrac{1}{2}\Delta\Delta^*, \tag{8.6}$$

$$\Xi Z^* = 0, \quad \operatorname{Im}(z_i^* \Xi_i) = 0, \quad \langle \Xi, R_n \rangle = 0,$$

$$d^2 = \|\Xi\|_F^2 + \|R_n\|_F^2, \quad \|R_n\|_F/d \le d/\sqrt{2N} \quad (d > 0). \tag{8.7}$$

Proof. The orbit is compact. Differentiating its squared distance at a minimum gives $\langle \Delta, AY\rangle = 0$ for every skew-Hermitian A and $\langle \Delta, iY\Phi\rangle = 0$ for every real diagonal Φ. The terms with Y replaced by Δ vanish, so ΔZ^* is Hermitian and $\mathrm{Im}(z_i^*\Delta_i) = 0$. Equal covariances give $\Delta Z^* + Z\Delta^* = -\Delta\Delta^*$. Multiplying the definition of Ξ by Z^* gives its horizontal constraint. Since C_n is Hermitian, its individual phase equations also hold. Horizontal arrays are orthogonal to C_nZ, and

$$\|R_n\|_F^2 = \frac{\mathrm{tr}((\Delta\Delta^*)^2)}{2N} \le \frac{d^4}{2N}.$$

Finally $\nabla\Gamma_K(Z) = 2KZS = 2N\lambda KZ$ proves stationarity, and $\Gamma_K(Z) = \mathrm{tr}(KZSZ^*) = N^2\lambda$. □

The simultaneous change of matrix is $A \mapsto D^*AD$: $(UYD)(D^*AD) = UYAD = 0$. Thus diagonal entries, rank and off-diagonal moduli are preserved. We choose the representative first and then fix its stress K; Γ_K need not be invariant under common unitaries when K is fixed. For equal covariances, $\Gamma_K(Y) - \Gamma_K(Z) = \mathcal{L}(Y) - \mathcal{L}(Z)$.

Split $\Xi = H + U$ into cluster means and zero-sum within-cluster residuals. This is an orthogonal splitting in the Frobenius metric. Both H and U satisfy the horizontal and individual phase equations. Every Hessian coefficient at Z depends only on its two cluster types; summing one residual factor gives zero. Consequently

$$Q(H, U) = Q(U, R_n) = 0. \tag{8.8}$$

8.2 The reference tensor and an endpoint perturbation bound

For a cluster-constant array, put $\|H\|_\nu^2 = \frac{1}{4}\sum_a \|H_a\|^2$. More generally give each member of cluster a weight $\nu_i = 1/(4m_a)$. The normalized reference potential is

$$\ell_0(Y) = \sum_{i,j} \nu_i\nu_j |y_i^* y_j| - \alpha \sum_i \nu_i \|y_i\|^2, \qquad B_0 = D^2\ell_0(F), \quad q_0 = B_0/2.$$

It has zero gradient at F: its phase matrix is $\sqrt{3}f_i^* f_j + (1-\sqrt{3})\mathbf{1}_{a(i)=a(j)}$. All infinitesimal common-unitary and individual-phase directions lie in the full kernel of B_0, including their pairing with ambient vectors.

Lemma 8.2 (Complete reference mean tensor)**.** *On*

$$\mathcal{H}_0 = \{H : \tfrac{1}{4}\sum_a H_a f_a^* = 0,\ \mathrm{Im}(f_a^* H_a) = 0\},$$

the full mean Hessian has eigenvalues $-2/\sqrt{3}$ with multiplicity three and $-\sqrt{3}$ with multiplicity two. On traceless Hermitian normal arrays MF, it equals $(2/\sqrt{3})I$; the scalar normal is null. The horizontal-to-normal operator norm is $\sqrt{2/3}$, and the full ambient mean operator norm is at most $7/4$.

Proof. Let v_a be tetrahedral Bloch vectors: $\sum_a v_a = 0$, $\sum_a v_a v_a^T = 4I_3/3$. Every phase-horizontal mean has the unique representation

$$H_a = a_a f_a + \tfrac{1}{2}(u_a \cdot \sigma) f_a, \qquad a_a \in \mathbb{R}, \quad u_a \perp v_a, \qquad \|H_a\|^2 = a_a^2 + \|u_a\|^2/4.$$

Here σ denotes the Pauli matrices. Put $A_1 = \sum_a a_a v_a$, $U_1 = \sum_a u_a$ and $D_{ab} = u_a \cdot v_b + v_a \cdot u_b$. Differentiating $r_a r_b \sqrt{(1 + x_a \cdot x_b)/2}$ at $r_a = 1$, $x_a = v_a$ gives

$$B_0(H, H) = -\frac{1}{2\sqrt{3}}\sum_a a_a^2 + \frac{(\sum_a a_a)^2}{8\sqrt{3}} + \frac{\sqrt{3}}{16} A_1 \cdot U_1 + \frac{\sqrt{3}}{32}\|U_1\|^2 - \frac{3\sqrt{3}}{256}\sum_{a\ne b} D_{ab}^2. \tag{8.9}$$

For verification, along a spherical curve with initial velocity u_a, the cross-pair square-root factor has derivatives $D_{ab}/(4h)$ and

$$\frac{u_a \cdot u_b + (\|u_a\|^2 + \|u_b\|^2)/6}{2h} - \frac{D_{ab}^2}{16h^3}, \qquad h = 1/\sqrt{3}.$$

Summing ordered pairs with weight $1/16$ gives (8.9); accelerations do not matter at a stationary point.

The matrix constraint is equivalent to $\sum a_a = 0$, $U_1 = -2A_1$, and $\sum u_a \times v_a = 0$. Set $b = 3A_1/4$. The component $a_a = b \cdot v_a$, $u_a = -(b - a_a v_a)$ has squared ν norm $\|b\|^2/2$ and Hessian $-\sqrt{3}\|b\|^2/3$. Its orthogonal complement has $a_a = 0$ and $U_1 = 0$. Writing $V = (v_a)$ and $U = (u_a)$, the latter is $U = TV$ with T symmetric, and $D = 2V^T TV$ gives $\|D\|_F^2 = (16/3)\|U\|_F^2$. In coordinates $v_a = (\pm 1, \pm 1, \pm 1)/\sqrt{3}$ with an even number of minus signs, $v_a^T T v_a = 0$ makes T diagonal and traceless. This gives the two shape eigenvalues $-\sqrt{3}$.

For $H_n = (b \cdot \sigma)F$, substitution and polarization give

$$\|H_n\|_\nu^2 = \|b\|^2, \quad B_0(H_n, H_n) = 2\|b\|^2/\sqrt{3}, \quad B_0(H, H_n) = -\sqrt{3}\, b \cdot A_1/4.$$

The horizontal constraints imply $\|H\|_\nu^2 \geq 9\|A_1\|^2/32$, with equality on the first sector. Thus the mixed norm is $\sqrt{2/3}$. Its coupled blocks have eigenvalues $\pm\sqrt{2}$, the shape eigenvalues have modulus $\sqrt{3}$, and all remaining gauge/scalar modes are zero. Hence the ambient norm is $\sqrt{3} < 7/4$. □

Lemma 8.3 (Squared reference projection)**.** *Suppose a mean array G satisfies*

$$\left\| \tfrac{1}{4} \sum_a G_a f_a^* \right\|_F \leq b\|G\|_\nu, \qquad |\operatorname{Im}(f_a^* G_a)| \leq \phi \|G_a\|.$$

Its reference-horizontal and Hermitian-normal components satisfy

$$\|G_0\|_\nu^2 \geq [1 - \phi^2 - 3(b + \phi)^2]\|G\|_\nu^2, \qquad \|G_n\|_\nu \leq \sqrt{2} b \|G\|_\nu.$$

Consequently, if $\beta = \phi^2 + 3(b + \phi)^2 < 1$,

$$B_0(G, G) \leq -g(b, \phi)\|G\|_\nu^2, \quad g(b, \phi) = \frac{2}{\sqrt{3}}(1 - \beta) - \frac{4}{\sqrt{3}}(b + b^2). \tag{8.10}$$

Proof. Remove the reference phase component, of norm at most $\phi\|G\|_\nu$. The remaining constraint matrix has norm at most $(b + \phi)\|G\|_\nu$. On phase-horizontal arrays the adjoint constraint is $M \mapsto P_{\mathrm{ph}}(M f_a)$, and for $M = A + iB$ with A, B Hermitian, $\operatorname{tr} B = 0$,

$$\|P_{\mathrm{ph}}(MF)\|_\nu^2 = \tfrac{1}{2}\|A\|_F^2 + \tfrac{1}{3}\|B\|_F^2.$$

Thus its least nonzero singular value is $1/\sqrt{3}$. The two successive corrections are orthogonal, proving the squared horizontal estimate. If $C = \frac{1}{4}\sum G_a f_a^*$, the normal projection is $2\operatorname{Herm}(C)F$ and has norm squared $2\|\operatorname{Herm}(C)\|_F^2$. Reference gauge directions are orthogonal to this normal and annihilated by B_0. Lemma 8.2, $\|G_0\|_\nu \leq \|G\|_\nu$, and the normal estimate yield (8.10). □

The following endpoint version of the modulus derivative estimate is useful because a certified overlap floor concerns actual overlaps, not the intervening line segment.

Lemma 8.4 (Endpoint Hessian bound)**.** *For nonzero $u, v \in \mathbb{C} \simeq \mathbb{R}^2$, and $0 < h \leq \min(|u|, |v|)$,*

$$\|D| \cdot |(u) - D| \cdot |(v)\| \leq |u - v|/h, \qquad \|D^2| \cdot |(u) - D^2| \cdot |(v)\| \leq \frac{7}{6}\,|u - v|/h^2.$$

Proof. The case $u = v$ is immediate. Write $r = |u| \geq s = |v|$, $t = r/s$, $d = t - 1$, $c = \cos\arg(u\bar{v})$, and $D = |u - v|/s = \sqrt{d^2 + 2t(1 - c)}$. The gradient difference has squared norm $(D^2 - d^2)/t \leq D^2$. The norm of the Hessian difference, multiplied by s, is exactly

$$\frac{d + \sqrt{d^2 + 4t(1 - c^2)}}{2t}.$$

Put $L = 7/6$ and $y = d/D$. Since $1 - c^2 \leq (D^2 - d^2)/t$,

$$tL^2D^2 - dLD - (1 - c^2) \geq \frac{D^2}{t}\left[(y - tL/2)^2 + \frac{3}{4}t^2L^2 - 1\right] > 0.$$

Also $2tLD \geq d$. Squaring in the correct direction proves the displayed Hessian norm is at most LD. Rescaling and $s \geq h$ prove the claim. No lower bound on intermediate segment moduli was assumed. □

For later use define, for a certified cross stress floor $0 < h_c \leq 15/26$,

$$\begin{aligned} d_p &= 2\eta + \eta^2 + \kappa(1 + \eta)^2, \quad u_p = \eta + \kappa(1 + \eta), \\ T_p &= \frac{7d_p}{6h_c^2} + \frac{u_p(2 + u_p)}{h_c} + \frac{d_p}{2h_c} + \frac{\kappa}{2}. \end{aligned} \tag{8.11}$$

Then the absolute *full* cross-pair Hessian error is at most $T_p(\|h_a\| + \|h_b\|)^2$. Indeed the overlap, first-variation and second-variation errors are bounded respectively by d_p, $u_p v$, and $\kappa v^2/2$, where $v = \|h_a\| + \|h_b\|$; the corresponding reference variations have bounds v and $v^2/2$. Apply Lemma 8.4 to the reference first-variation arguments, then change those arguments, the gradient, and the second variation. This gives precisely the four terms in (8.11). Same-type *mean* pairs have exact Hessian $2h_a^* K h_a$, and hence error at most $2\kappa\|h_a\|^2$.

8.3 Mean curvature in physical, mass, and stress coordinates

An ambient physical mean perturbation bound needed for the normal terms is

$$\begin{aligned} W_\times &= \max_{a \neq b} |16p_a p_b - 1|, \qquad W_d = 8e + 16e^2, \\ E_{\mathrm{Hess}} &= (1 + 4e)^2(3T_p + \kappa/2) + (27/4)W_\times + W_d/2 \\ &\quad + 2\{(1 + 4e)[(7/10 + \tau)\kappa + \tau] + (14/5)e\}. \end{aligned} \tag{8.12}$$

It satisfies

$$|N^{-2}Q(H, J) - q_0(H, J)| \leq (E_{\mathrm{Hess}}/2)\|H\|_\nu \|J\|_\nu \tag{8.13}$$

on all cluster-constant arrays, without a horizontal hypothesis. To verify this, use $\sum_{a \neq b}(l_a + l_b)^2/16 \leq 3\|H\|_\nu^2$. The reference cross Hessian is bounded by $(9/4)(l_a + l_b)^2$; the same-type one is exactly $2l_a^2$. This gives the cross and diagonal weight errors separately. The multiplier error follows from $\|\lambda K - \alpha I\| \leq (7/10 + \tau)\kappa + \tau$ and $|4p_a - 1| \leq 4e$. Polarization of the resulting absolute self-adjoint operator bound gives (8.13). For the four profiles, respectively, valid cross-weight envelopes are

$$W_\times \leq 2/N + 3/N^2, \quad 6/N + 15/N^2, \quad 4/N + 4/N^2, \quad 4/N + 12/N^2.$$

They follow directly from $16p_a p_b - 1 = 4(\delta_a + \delta_b)/N + 16\delta_a\delta_b/N^2$.

Lemma 8.5 (Mass coordinates)**.** *Let $\epsilon_+ \geq \max_a(p_a - 1/4)$ cover both signs of the count profile, and define*

$$b_g = \sqrt{B\sigma^2/\omega_p}, \quad g_g = g(b_g, \eta) - (1 + 2/\omega_p)T_p - \kappa/2 - 2(\alpha\kappa + \tau)/\omega_p - 8\alpha\epsilon_+/\omega_p.$$

If the reference projection fraction is positive and $g_g > 0$, then

$$Q(H) \le -Nc_g\|H\|_F^2, \qquad c_g = \omega_p g_g/2 \tag{8.14}$$

for every actual horizontal mean.

Proof. Set $G_a = 4p_aH_a$. Then $\frac{1}{4}\sum G_a z_a^* = 0$ and $\operatorname{Im}(z_a^*G_a) = 0$. The reference constraint error is at most $b_g\|G\|_\nu$, and its phase error is at most η. Lemma 8.3 gives $g(b_g,\eta)$. The exact norm relation is $\|H\|_F^2/N = \frac{1}{4}\sum\|G_a\|^2/(4p_a) \le \|G\|_\nu^2/\omega_p$.

For completeness, the mass cancellation can be checked at the half-Hessian level. Put $\theta_{ab}^0 = f_a^*f_b/|f_a^*f_b|$ and $A_{ab}(g) = [\operatorname{Im}(\overline{\theta_{ab}^0}g^*f_b)]^2/|f_a^*f_b|$. If q_p^F denotes the reference half-Hessian with masses p, then

$$q_p^F(H) - q_{1/4}^F(G) = \sum_a \frac{\sum_b (p_b - p_a)A_{ab}(G_a) + 4\alpha(p_a - \frac{1}{4})\|G_a\|^2}{4(4p_a)}.$$

Mixed first-variation and second-variation terms cancel because $p_ap_b/[(4p_a)(4p_b)] = 1/16$. In coordinates $f_a = (1,0)$, $g = (x+iy, u+iv)$, $\sum_b A_{ab}(g) = (1+\sqrt{3})y^2 + \sqrt{3}(u^2+v^2) \le 4\alpha\|g\|^2$. The displayed difference is therefore at most $4\alpha\epsilon_+\|G\|_\nu^2/\omega_p$.

Moreover

$$\sum_{a\ne b} p_ap_b(\|H_a\| + \|H_b\|)^2 \le (1 + 2/\omega_p)\|G\|_\nu^2.$$

Expanding the two squared-column terms gives at most $(4/\omega_p - 1)\sum\|G_a\|^2/8$, and the product terms at most $3\sum\|G_a\|^2/8$. The same-type full-Hessian parameter error is $\kappa\|G\|_\nu^2/2$, and the covariance error is $2(\alpha\kappa+\tau)\|G\|_\nu^2/\omega_p$. These estimates give g_g. Its strict positivity is checked before converting the negative norm estimate. □

Lemma 8.6 (Stress and mass coordinates)**.** *Put*

$$d_K = \frac{\kappa}{1+\sqrt{1-\kappa}}, \qquad \eta_v = \sqrt{1+\kappa}\,\eta + d_K, \qquad b_v = \frac{\sqrt{B\sigma^2} + d_K}{\sqrt{\omega_p}},$$
$$\phi_v = \eta_v + \frac{\kappa(1+\eta)}{\sqrt{1-\kappa}}, \qquad d_v = 2\eta_v + \eta_v^2,$$
$$T_v = \frac{7d_v}{6h_c^2} + \frac{\eta_v(2+\eta_v)}{h_c} + \frac{d_v}{2h_c},$$
$$g_v = g(b_v,\phi_v) - (1+2/\omega_p)T_v - 2\tau/\omega_p - 8\alpha\epsilon_+/\omega_p.$$

Assume $\eta_v < 1/32$, $\phi_v^2 + 3(b_v+\phi_v)^2 < 1$ and $g_v > 0$. Then

$$Q(H) \le -Nc_v\|H\|_F^2, \qquad c_v = \omega_p(1-\kappa)g_v/2. \tag{8.15}$$

Proof. Set $T = K^{1/2}$, $v_a = Tz_a$, and $G_a = 4p_aTH_a$. The entire Lagrangian in TY has pair stress I and multiplier λI. The transformed covariance is $K/2$, not $I/2$; no tightness of v is used. Exactly $\frac{1}{4}\sum G_av_a^* = 0$, but its phase equation has the defect

$$|\operatorname{Im}(v_a^*G_a)| = 4p_a|\operatorname{Im}(z_a^*(K-I)H_a)| \le \frac{\kappa(1+\eta)}{\sqrt{1-\kappa}}\|G_a\|.$$

Spectral calculus gives $\|T - I\| \le d_K$, proving the point bound $\|v_a - f_a\| \le \eta_v$. For the aggregate use $v - f = (z-f) + (T-I)z$ and the *original* identity $\sum p_a\|z_a\|^2 = 1$:

$$\Big(\sum p_a\|v_a - f_a\|^2\Big)^{1/2} \le \sqrt{B\sigma^2} + d_K.$$

Thus Lemma 8.3 applies with b_v, ϕ_v; in particular the transformed phase defect has not been erased.

The cross-pair perturbation is now (8.11) with point error η_v and stress error zero, namely T_v. The overlap floor is unchanged, since $v_a^* v_b = z_a^* K z_b$. Same-type parameter error is exactly zero, because both kernels have Hessian $2\|g_a\|^2$. Their mass error is still included in the preceding mass-coordinate identity. The covariance error is $2\tau/\omega_p$. This proves the Hessian bound with coefficient g_v. Finally

$$\|G\|_\nu^2 = \sum_a 4p_a^2 H_a^* K H_a \geq \omega_p(1-\kappa)\|H\|_F^2/N.$$

Its substitution into a negative bound is valid after $g_v > 0$. The transformation is invertible and introduces no equality mode. □

The accepted mean coefficient may be any certified lower bound for $\max(c_g, c_v)$, using only those alternatives whose guards hold. The stress transform is confined to this mean estimate. In particular neither b_g nor b_v replaces the physical normal projection below.

8.4 The intrinsic residual bound

We use the two-dimensional spectral reduction, since a trace-only bound would lose the anisotropy needed in the nonuniform profiles. Use the candidate construction of Section 6, writing its auxiliary rank-two Gram matrix as $\mathcal{B}$ to distinguish it from the aggregate coefficient B in (8.3). Define

$$\ell_a = \frac{1-\lambda-p_a}{(1-\lambda)(\lambda-2p_a)}, \quad L = \operatorname{diag}(\ell_a), \quad \mathcal{T} = L - L\mathcal{B}L, \quad \mathcal{T}_0 = \operatorname{tr}\mathcal{T},$$
$$d_a = \ell_a[1-\ell_a(\lambda-p_a)], \quad r_{Ka}^2 = \frac{d_a}{p_a\mathcal{T}_0}, \qquad s_{ab} = \frac{\ell_a\ell_b}{\mathcal{T}_0} \quad (a\neq b), \quad s_{aa} = r_{Ka}^2. \tag{8.16}$$

All displayed denominators and d_a are strictly positive. If the columns of U form an orthonormal basis of range $\mathcal{T}$, then $K = 2U^*\mathcal{T}U/\mathcal{T}_0$ and $z_a = U^* e_a/\sqrt{2p_a}$. The exact identity $\mathcal{T}L^{-1}\mathcal{T} = \mathcal{T}$ implies, for $v_a = K^{1/2} z_a$,

$$\sum_a (p_a/\ell_a) v_a v_a^* = I_2/\mathcal{T}_0. \tag{8.17}$$

Indeed $U^* L^{-1} U = (U^*\mathcal{T}U)^{-1}$.

Lemma 8.7 (Pointwise intrinsic form)**.** *For a physical vector h at type a define*

$$q_a(h) = \sum_b \frac{p_b}{s_{ba}}[\operatorname{Im}(\overline{\theta_{ba}} z_b^* K h)]^2 - \lambda h^* K h, \qquad \theta_{ba} = z_b^* K z_a / s_{ba}.$$

For the full within-cluster zero-sum residual U, $Q(U)/N = \sum_j q_{a(j)}(U_j)$. For $j \neq a$ put

$$u_j = p_j^2(r_{Kj}^2 - s_{aj}^2/r_{Ka}^2), \qquad D_a = \frac{2\sum_{j<k} u_j u_k - \sum_j u_j^2}{4},$$

where the sums run over the three indices different from a, and set

$$\mathcal{D}_a = D_a \sum_{\substack{j<k \\ j,k\neq a}} \frac{1}{p_j p_k s_{aj} s_{ak}}, \qquad c_{Ka} = \lambda - \frac{\ell_a^{-1} + \sqrt{\ell_a^{-2} - 4\mathcal{D}_a}}{2}. \tag{8.18}$$

If $c_K \leq \min_a\{\lambda, c_{Ka}\}$ is strictly positive, then

$$q_a(h) \leq -(1-\kappa)c_K\|h\|^2 \quad \textit{whenever } \operatorname{Im}(z_a^* h) = 0. \tag{8.19}$$

Proof. In the ordered Hessian, mixed-argument and mixed first-variation terms of a zero-sum residual factor through its group sums and vanish. The two pure-first terms give $Q(U)/N = \sum_j q_{a(j)}(U_j)$, including same-cluster and diagonal pairs.

Fix $e_0 = v_a/r_{Ka}$ and a perpendicular unit vector e_1. Rotate v_b into its pair phase, writing $(s_{ab}/r_{Ka})e_0 + c_b e_1$, with $c_a = 0$. Stress stationarity gives $\sum_b p_b c_b = 0$ and $\sum_b p_b s_{ab} = \lambda r_{Ka}^2$. If $K^{1/2}h = (x+iy)e_0 + ue_1$, direct expansion therefore yields

$$q_a(h) = -\lambda x^2 + \sum_{b\neq a} \frac{p_b}{s_{ab}} [\mathrm{Im}(\bar{u}c_b)]^2 - \lambda|u|^2.$$

All terms containing y cancel exactly; the whole iz_a direction is in the kernel, including its bilinear pairings.

Let M_a be the real two-dimensional positive quadratic form in the middle sum. Each real rank-one form $[\mathrm{Im}(\bar{u}c_b)]^2$ has trace $|c_b|^2$. Since $c_a = 0$ and $s_{ab} = \ell_a\ell_b/\mathcal{T}_0$ for $b \neq a$,

$$\mathrm{tr}\, M_a = \sum_{b\neq a} \frac{p_b|c_b|^2}{s_{ab}} = \frac{\mathcal{T}_0}{\ell_a} \sum_b \frac{p_b}{\ell_b}|c_b|^2,$$

$$\sum_b \frac{p_b}{\ell_b}|c_b|^2 = e_1^* \left(\sum_b \frac{p_b}{\ell_b} v_b v_b^* \right) e_1 = \frac{1}{\mathcal{T}_0}.$$

The last equality is (8.17); pair-phase rotations do not change the projected moduli. Thus $\mathrm{tr}\, M_a = 1/\ell_a$. The vectors $t_j = p_j c_j$ form a closed triangle. Their pairwise determinants have common square D_a, since $t_j \cdot t_k = (u_l - u_j - u_k)/2$. The two-dimensional Cauchy–Binet formula gives $\det M_a = \mathcal{D}_a$. This proves its *complete* largest-eigenvalue formula (8.18).

For a physical phase-horizontal h, choose real t so that $h' = h - itz_a$ satisfies $\mathrm{Im}(z_a^* K h') = 0$. Then $q_a(h') = q_a(h)$, while $\|h'\|^2 = \|h\|^2 + t^2\|z_a\|^2 \geq \|h\|^2$. The spectral bound and $K \geq (1-\kappa)I$ prove (8.19). This is a pointwise statement, prior to imposing any zero group sum. The phase adjustment is real-linear within each group, so it would also preserve such sums if applied to an array. □

In the accepted comparisons we use a directed lower bound

$$c_r = (1-\kappa)c_K > 0 \tag{8.20}$$

from this intrinsic lemma. All residual lower bounds used in the certificates follow from this lemma. The scalar enclosures cover the *whole* reciprocal-N interval, not merely the local endpoint. They are produced by (8.16)–(8.18) with the root enclosure and contractor described next.

Write $x = 1/N$, $p_a = 1/4 + \delta_a x$, $h_0 = \alpha - 1/2$, $\lambda = \alpha - \zeta$, and $h = h_0 - \zeta$. Polynomial division of the defining candidate equation gives exactly

$$\zeta = x^2 \frac{3 + 1/h - 1/(4h^2)}{2 + 1/(4h_0 h)} \sum_a \frac{\delta_a^2}{h - 2\delta_a x}. \tag{8.21}$$

For verification, $\sum_a (h - 2\delta_a x)^{-1} = 4/h + (4x^2/h^2)\sum_a \delta_a^2/(h - 2\delta_a x)$; the uniform equation is $2h_0 + 1 - 1/(4h_0) = 0$. On any closed x interval start with $0 \leq \zeta \leq (2/3)Cx_{\mathrm{hi}}^2$ and intersect this interval three times with the outward interval image of (8.21). The true root remains inside each intersection; convergence is not assumed. Positivity of every denominator is checked before division. For the actual candidate the discriminant is nonnegative. Its enclosing interval may therefore be intersected with $[0, \infty)$; a negative lower endpoint is replaced by zero, while a negative upper endpoint fails the check. Taking the minimum over all intervals, signs and four vertex types gives c_K. All intervals meeting a requested endpoint are retained whole, including any interval crossing it.

8.5 The normal component and its finite budget

The candidate diagonal identity is $p_a\|z_a\|^2 = 1/4 + (p_a - 1/4)/b$ with $b = 3(2\lambda - 1) > 109/100$. It implies

$$\tfrac{1}{4}\sum_a \|f_a - 4p_a z_a\|^2 = \sum_a p_a\|z_a - f_a\|^2 + 4\sigma^2/b.$$

Set

$$b_{\mathrm{ph}} = \sqrt{(B + 400/109)\sigma^2}.$$

For a physical horizontal mean H this bounds its reference constraint matrix by $b_{\mathrm{ph}}\|H\|_\nu$; in particular its reference normal projection has norm at most $\sqrt{2}b_{\mathrm{ph}}\|H\|_\nu$. The following lemma uses these bounds in the original frame variables.

Lemma 8.8 (Positive normal coupling)**.** *Define*

$$t = \min\{\sqrt{2}\eta, \sqrt{2B\sigma^2/\omega_p}\}, \qquad a = \min\{\omega_p^{-1/2}, 1 + t\},$$

and

$$\begin{aligned} J_* &= \frac{(1/2 + b_{\mathrm{ph}})/\sqrt{3} + (7/8)t + (E_{\mathrm{Hess}}/2)a}{\sqrt{\omega_p}}, \\ L'_* &= \frac{1}{2\sqrt{3}} + \frac{7}{4}t + \frac{7}{8}t^2 + (E_{\mathrm{Hess}}/2)a^2, \\ H_{\mathrm{amb}} &= (7/4 + E_{\mathrm{Hess}})/(2\omega_p), \qquad J = \min(J_*, H_{\mathrm{amb}}), \quad L = \min(L'_*, H_{\mathrm{amb}}). \end{aligned} \tag{8.22}$$

Then $|Q(H, R_n)| \le NJ\|H\|_F\|R_n\|_F$ *and* $Q(R_n) \le NL\|R_n\|_F^2$.

Proof. Write $C_n = cI + D$ with $\operatorname{tr} D = 0$ and set $r = \|R_n\|_F/\sqrt{N}$. Tightness gives $r^2 = \operatorname{tr}(C_n^2)/2 = \|C_nF\|_\nu^2$. Because $C_n \succeq 0$ in dimension two, $\|DF\|_\nu^2 = \operatorname{tr}(D^2)/2 \le r^2/2$. This half-energy estimate would fail for an indefinite normal. Also $\|C_n\| \le \sqrt{2}r$, so for $E = C_n(Z - F)$, $\|E\|_\nu \le tr$, and $\|R_n\|_\nu \le ar$. Both the point and aggregate alternatives in t refer to this same E.

By the reference tensor and the physical normal projection,

$$|q_0(H, C_nF)| \le \frac{1/2 + b_{\mathrm{ph}}}{\sqrt{3}}\|H\|_\nu r.$$

The error $q_0(H, E)$ is at most $(7/8)t\|H\|_\nu r$. Add (8.13) and use $\|H\|_\nu \le \|H\|_F/\sqrt{N\omega_p}$ to obtain J_*. For the pure normal, the scalar part is null and the traceless part contributes at most $r^2/(2\sqrt{3})$ to $q_0(C_nF)$. The two terms involving E are bounded by $(7t/4 + 7t^2/8)r^2$, and the perturbation contributes $E_{\mathrm{Hess}}a^2r^2/2$. This gives L'_*. The ambient operator bound independently gives H_{amb} for each. □

Let $c_m > 0$ be the accepted mean lower bound and $c_r > 0$ the intrinsic residual lower bound. Put $h = \|H\|_F/d$, $r = \|U\|_F/d$, $z = \|R_n\|_F/d$. Although $\Delta = \Xi - R_n$, an absolute mixed upper bound gives

$$\frac{Q(\Delta)}{Nd^2} \le -c_m h^2 - c_r r^2 + 2Jzh + Lz^2, \qquad h^2 + r^2 + z^2 = 1, \quad 0 \le z \le \rho_n. \tag{8.23}$$

If $d \le d_c/\sqrt{N}$, take $\rho_n^2 = d_c^2/(2N^2) \le 1/2$. The exact supremum of the right-hand side is

$$\begin{cases} -c_r + (c_r + L + J^2/(c_m - c_r))\rho_n^2, & c_m > c_r,\ J^2\rho_n^2 \le (c_m - c_r)^2(1 - \rho_n^2), \\ -c_m + (c_m + L)\rho_n^2 + 2J\sqrt{\rho_n^2(1 - \rho_n^2)}, & \text{otherwise.} \end{cases} \tag{8.24}$$

To prove this, fix z and maximize $-(c_m - c_r)h^2 + 2Jzh$ on $[0, \sqrt{1 - z^2}]$. The interior value and the boundary value are nondecreasing in z for $z \le 1/\sqrt{2}$, and agree at the transition. The boundary case includes $c_m \le c_r$, so no division by a nonpositive gap occurs. Denote the negative of (8.24) by g_{normal}. All branch choices are made with the conservative rational coefficients actually supplied to the certificate.

8.6 Angular transport and the shared radial–tangent budget

The columnwise tangent used in the Taylor remainder is different from the horizontal array Ξ used in the Hessian. Define

$$D_r = \frac{(4/11)e}{1-4e}, \qquad r_- = \sqrt{1-D_r}, \quad r_+ = \sqrt{1+D_r}, \qquad d_c = \sqrt{A_R} + \sqrt{BC}.$$

The candidate radial identity in Section 6 gives $r_- \le \|z_a\| \le r_+$. Closest alignment and the triangle inequality give $d \le d_c/\sqrt{N}$. In the preliminary common alignment, the exact identity

$$\|rv - f\|^2 = (r-1)^2 + r\, d_{\mathrm{ph}}(v,f)^2$$

for unit v, f in closest phases gives candidate angular bounds $BC/(Nr_-)$ in sum and $C_{\mathrm{pt}}C/(N^2 r_-)$ pointwise. Put

$$B_{\mathrm{ang}} = \sqrt{L_U \bar{E}} + \sqrt{BC/r_-}, \qquad \chi = \sqrt{\frac{(1+s)r_+}{(1-s)r_-}}, \qquad D_U = \frac{2}{3} - \frac{L_*}{N} > 0.$$

For fixed common unitary, minimizing individual phases gives

$$\|y_i - z_i\|^2 = (\|y_i\| - \|z_i\|)^2 + \|y_i\|\|z_i\| d_{\mathrm{ph}}(u_i, z_i/\|z_i\|)^2.$$

Therefore the actual closest unitary minimizes the angular sum with the positive weights $\|y_i\|\|z_i\|$. Comparison to the preliminary alignment yields

$$\sum_i \|y_i\|\|z_i\| d_{\mathrm{ph}}(u_i, z_i/\|z_i\|)^2 \le (1+s) r_+ B_{\mathrm{ang}}^2/N, \tag{8.25}$$

$$\sum_i d_{\mathrm{ph}}(u_i, z_i/\|z_i\|)^2 \le \chi^2 B_{\mathrm{ang}}^2/N.$$

No unweighted angular optimality is used.

For completeness, the rotation estimate needed to transfer the pointwise bound follows from the tetrahedral second moment. Modulo scalar phase write $U = \cos\theta I + i \sin\theta (n\cdot\sigma)$ and put $t = |\sin\theta|$. If v_a are the reference Bloch vectors, then $|f_a^* U f_a|^2 = 1 - t^2(1-(n\cdot v_a)^2)$ and $d_{\mathrm{ph}}(Uf_a, f_a)^2 \ge t^2(1-(n\cdot v_a)^2)$. Since $\sum v_a v_a^T = 4I/3$,

$$\sum_i d_{\mathrm{ph}}(U f_{a(i)}, f_{a(i)})^2 \ge N D_U t^2.$$

The phase-distance triangle inequality and (8.25) bound its left side by $(1+\chi)^2 B_{\mathrm{ang}}^2/N$. If

$$\frac{(1+\chi)^2 B_{\mathrm{ang}}^2}{N^2 D_U} < 1 - (255/256)^2, \tag{8.26}$$

every $|f_a^* U f_a| > 255/256$, and

$$d_{\mathrm{ph}}(Uf_a, f_a)^2 = \frac{2(1-|f_a^* U f_a|^2)}{1+|f_a^* U f_a|} \le \frac{512}{511} t^2.$$

Thus one may use

$$\mathrm{Rot}_{\mathrm{ang}} = \sqrt{\frac{512(1+\chi)^2 B_{\mathrm{ang}}^2}{511 D_U}}.$$

An independent physical-distance triangle gives $\sum_i d_{\mathrm{ph}}(Uf_a, f_a)^2 \le 4d_c^2/N$ and hence $\mathrm{Rot}_{\mathrm{phys}} = \sqrt{(512/511)4d_c^2/D_U}$, provided $4d_c^2/(N^2 D_U) < 1 - (255/256)^2$. Use the smaller of these two bounds only when each selected bound has its own stated guard; denote the result by Rot.

Write, in the actual individual phases,

$$y_i = (1+t_i)z_i + \xi_i = r_i z_i + \xi_i, \qquad z_i^* \xi_i = 0, \quad t_i \in \mathbb{R}. \tag{8.27}$$

Let $X^2 = \sum_i \|\xi_i\|^2$ and let T bound $\max_i \|\xi_i\|$. The following simultaneous choices are valid:

$$\begin{aligned} q_t &= \frac{1}{N}\min\{d_c, (1+s)\sqrt{r_+/r_-}B_{\mathrm{ang}}\}, & X/\sqrt{N} &\le q_t, \\ \epsilon &= \sqrt{P_R/N} + (\mathrm{Rot}_{\mathrm{phys}} + \sqrt{C_{\mathrm{pt}}C})/N, & q_d &= d_c/N, \\ T &= \min\left\{\epsilon, (1+s)\left[\sqrt{L_P\bar{E}/N} + \frac{\mathrm{Rot}}{N} + \frac{\sqrt{C_{\mathrm{pt}}C}}{N\sqrt{r_-}}\right]\right\}. \end{aligned} \tag{8.28}$$

Indeed $1 - |u^*v|^2 \le d_{\mathrm{ph}}(u,v)^2$ for unit vectors. Multiplying (8.25) by $(1+s)/r_-$ gives $X^2 \le (1+s)^2(r_+/r_-)B_{\mathrm{ang}}^2/N$. The pointwise estimate follows from the same triangle inequality with the rotation bound, while the physical triangle gives ϵ. An angular displacement has therefore not been charged a first-order radial point error.

The certificate checks $T < 1 - s$. Since $z_i^* y_i \ge 0$ in closest phases,

$$0 < \rho = \frac{\sqrt{(1-s)^2 - T^2}}{r_+} \le r_i \le R = \frac{1+s}{r_-}. \tag{8.29}$$

Orthogonality in (8.27) gives the exact identity

$$d^2 = \sum_i t_i^2 \|z_i\|^2 + X^2, \qquad T_r^2 := \sum_i t_i^2 \le (d^2 - X^2)/r_-^2. \tag{8.30}$$

This budget is distinct from the global radial, moment and angular deficits. Equations (8.28) are the explicit transfer between them. For $d > 0$ put $\vartheta = X/d \in [0,1]$.

8.7 The pair remainder and physical fiber bounds

For an ordered pair put $s_{ij} = |z_i^* K z_j|$, and rotate its candidate overlap to the positive real axis. In this phase set

$$\alpha_{ij} = \overline{S}_{ij} z_i^* K \xi_j, \quad \beta_{ij} = \overline{S}_{ij} \xi_i^* K z_j, \quad \gamma_{ij} = \overline{S}_{ij} \xi_i^* K \xi_j, \qquad (a_{ij}, b_{ij}, c_{ij}) = \mathrm{Im}(\alpha_{ij}, \beta_{ij}, \gamma_{ij}).$$

For this paragraph abbreviate $s = s_{ij}$, $p = r_i r_j$ and $L = r_i\alpha_{ij} + r_j\beta_{ij}$. The complete Taylor polynomial through total degree two in (t, ξ) is

$$P_2 = ps + \mathrm{Re}(L + \gamma_{ij}) + (a_{ij} + b_{ij})^2/(2s).$$

With $R_s(h) = |s+h| - s - \mathrm{Re}\, h - (\mathrm{Im}\, h)^2/(2s)$, the exact remainder is

$$\begin{aligned} |ps + L + \gamma_{ij}| - P_2 = \frac{r_i - r_j}{2s}\left(\frac{a_{ij}^2}{r_j} - \frac{b_{ij}^2}{r_i}\right) + \frac{(r_i a_{ij} + r_j b_{ij})c_{ij}}{ps} + \frac{c_{ij}^2}{2ps} \\ + pR_s\left(\frac{\alpha_{ij}}{r_j} + \frac{\beta_{ij}}{r_i} + \frac{\gamma_{ij}}{p}\right). \end{aligned} \tag{8.31}$$

To verify it, expand the square of $(r_i a_{ij} + r_j b_{ij} + c_{ij})$ divided by $2ps$; the $a_{ij}b_{ij}$ term cancels exactly. In particular the pure radial dependence is exactly quadratic. All within-cluster and diagonal pairs remain in this identity.

For types c (different clusters) and s (same cluster), choose positive floors h_t and upper bounds m_t, g_t such that

$$s_{ij} \ge h_t, \quad |\alpha_{ij}| \le m_t\|\xi_j\|, \quad |\beta_{ij}| \le m_t\|\xi_i\|, \quad |\gamma_{ij}| \le g_t\|\xi_i\|\|\xi_j\|. \tag{8.32}$$

We use $h_{\mathrm{c}} = h_c$ for the certified cross floor. Besides the certified candidate overlap floor, valid generic choices are

$$\begin{aligned} h_E &= 15/26 - 2\eta - \eta^2 > 0, \qquad h_{\mathrm{s}} = (1-\kappa) r_-^2, \\ m_{\mathrm{c}} &= \sqrt{r_+^2 - h_E^2/r_+^2} + \kappa r_+, \qquad m_{\mathrm{s}} = \kappa r_+, \\ g_{\mathrm{c}} &= \kappa + \min\{1, (1/\sqrt{3} + 2\eta + \eta^2)/r_-^2\}, \qquad g_{\mathrm{s}} = 1 + \kappa. \end{aligned} \tag{8.33}$$

The displayed choice of h_{s} may be replaced by another independently proved floor; in all cases require $0 < h_{\mathrm{s}} \le \min_a s_{aa}$. The projected-linear bound follows from the Euclidean cross floor and the component of $K - I$. For the quadratic bound, define the antiunitary $J(v_1, v_2) = (-\overline{v}_2, \overline{v}_1)$. Each $\xi_i = c_i J(z_i/\|z_i\|)$, so

$$|\xi_i^* \xi_j| = \frac{|z_i^* z_j|}{\|z_i\| \|z_j\|} \|\xi_i\| \|\xi_j\|.$$

Adding the $K - I$ contribution proves $g_{\mathrm{c}}, g_{\mathrm{s}}$. In particular this is a physical complex-line calculation, valid for every within-cluster residual.

8.8 Row estimates coupled to intrinsic residual curvature

Write $e_i = \|\xi_i\|$. Retain the separate pointwise coefficient

$$M = \max\{m_{\mathrm{c}}^2/h_{\mathrm{c}}, m_{\mathrm{s}}^2/h_{\mathrm{s}}\}.$$

The following row bounds keep the complete imaginary parts in (8.31):

$$\sum_i a_{ij}^2/s_{ij} \le N H e_j^2, \qquad \sum_{a(i)\neq a(j)} a_{ij}^2/s_{ij} \le N H_{\mathrm{c}} e_j^2.$$

To obtain an initial bound, put

$$\begin{aligned} d_g = 2\eta + \eta^2 + \kappa r_+^2, \qquad \delta &= \eta + \kappa r_+ + \eta/\sqrt{r_-} + \sqrt{2/3}\, d_g/\sqrt{h_{\mathrm{c}}/\sqrt{3}}, \\ H_{\mathrm{iso}} &= \frac{(\sqrt{p_+} + \sqrt{1-p_-}\,\delta)^2}{h_{\mathrm{c}}}, \\ H = \min\{M, H_{\mathrm{iso}} + p_+ m_{\mathrm{s}}^2/h_{\mathrm{s}}\}, \qquad H_{\mathrm{c}} &= \min\{H_{\mathrm{iso}}, (1-p_-) m_{\mathrm{c}}^2/h_{\mathrm{c}}\}. \end{aligned} \tag{8.34}$$

The subscript on H_{iso} denotes the reference isotropy estimate. For a reference center f_b, the three projected cross forms have amplitude $\sqrt{2/3}\|\xi\|$ and phases separated by $2\pi/3$. Their squared imaginary parts sum to $\|\xi\|^2$, so their weighted sum is at most $p_+\|\xi\|^2$. Transport $\xi = cJ(z_b/\|z_b\|)$ to cJf_b; its change is at most $\eta\|\xi\|/\sqrt{r_-}$. For complex nonzero v, v_0,

$$\left| \frac{v}{|v|} - \frac{v_0}{|v_0|} \right|^2 = \frac{|v - v_0|^2 - (|v| - |v_0|)^2}{|v||v_0|}.$$

The overlap phase changes by at most $d_g/\sqrt{h_{\mathrm{c}}/\sqrt{3}}$. These bounds give the displayed δ for the full phase-aligned linear form. The weighted Euclidean triangle inequality proves H_{iso}; same-cluster rows and pointwise M give the other terms.

There is a second, sharper row bound directly from intrinsic curvature. Let c_{intr} be the physical pointwise coefficient established in the residual argument above, before imposing any zero-sum condition:

$$\sum_b p_b \frac{[\mathrm{Im}(\overline{S}_{ba} z_b^* K h)]^2}{|z_b^* K z_a|} - \lambda h^* K h \le -c_{\mathrm{intr}} \|h\|^2, \qquad \mathrm{Im}(z_a^* h) = 0. \tag{8.35}$$

Every columnwise ξ_j is eligible since $z_j^*\xi_j = 0$. Therefore, for any certified upper λ_+,

$$H_{\mathrm{spec}} = \lambda_+(1+\kappa) - c_{\mathrm{intr}}, \qquad H \leftarrow \min(H, H_{\mathrm{spec}}), \quad H_{\mathrm{c}} \leftarrow \min(H_{\mathrm{c}}, H_{\mathrm{spec}}). \tag{8.36}$$

The certificate checks $H_{\mathrm{spec}} \geq 0$. Every summand in the row is nonnegative, which justifies the cross-only bound. The coefficient M is not changed: a row bound does not imply a pointwise pair bound. Nor is a mean curvature coefficient substituted for c_{intr}. Using (8.35) both here and in quadratic descent is valid because both are pointwise consequences of the same inequality; neither spends a separate portion of a deficit.

Hermitian symmetry gives $b_{ij} = -a_{ji}$, $c_{ji} = -c_{ij}$. The full ordered radial term in (8.31) is consequently

$$\sum_{i,j} (t_i - t_j) a_{ij}^2/(r_j s_{ij}).$$

Put $W_i = \sum_j a_{ij}^2/s_{ij}$. Then $0 \leq W_i \leq MX^2$, $\sum_i W_i \leq NHX^2$, hence $\sum_i W_i^2 \leq NMHX^4$. Cauchy–Schwarz, followed by (8.30), bounds this term divided by Nd^2 by

$$B_{\mathrm{rad}}\vartheta\sqrt{1-\vartheta^2}, \qquad B_{\mathrm{rad}} = \frac{HT + \sqrt{MH}\, q_t}{\rho r_-}. \tag{8.37}$$

Indeed the two sums with $|t_i|$ and $|t_j|$ are at most $\sqrt{NMH}\, T_r X^2$ and $NHT_r TX$, respectively, before division by ρ.

The full ordered linear–quadratic term is $2\sum_{i,j} a_{ij}c_{ij}/(r_j s_{ij})$. If $E_b = \sum_{a(j)=b} e_j^2$, its cross part, by weighted Cauchy–Schwarz, is at most

$$\frac{2g_{\mathrm{c}}}{\rho}\sqrt{NH_{\mathrm{c}}/h_{\mathrm{c}}}\sum_b E_b\sqrt{X^2 - E_b} \leq \frac{\sqrt{3}g_{\mathrm{c}}}{\rho}\sqrt{NH_{\mathrm{c}}/h_{\mathrm{c}}}X^3.$$

The last inequality follows from $\sum E_b^2 \geq X^4/4$. The same-cluster part is at most $2\sqrt{Np_+}m_{\mathrm{s}}g_{\mathrm{s}}X^3/(\rho h_{\mathrm{s}})$. Set

$$L_{\mathrm{row}} = \frac{\sqrt{3}}{2}g_{\mathrm{c}}\sqrt{H_{\mathrm{c}}/h_{\mathrm{c}}} + \sqrt{p_+}m_{\mathrm{s}}g_{\mathrm{s}}/h_{\mathrm{s}}, \qquad J_{\max} = \max_t g_t^2/(2h_t),$$
$$A_{\mathrm{mixed}} = 2L_{\mathrm{mixed}}q_t/\rho + J_{\max}q_t^2/\rho^2, \tag{8.38}$$

where L_{mixed} can be L_{row} or its minimum with the independent weighted type bound described next. The second term in A_{mixed} retains the full quadratic–quadratic term c_{ij}^2. The same-cluster term in L_{row} is also retained.

8.9 Complete scalar bounds and four-cluster summation

The elementary modulus identities below retain all orders of the remainder; no assertion that its argument has modulus less than its base is required. For $h \neq 0$ and $s > 0$ put

$$z = |h|/s, \qquad t = (1 - |1 + h/s|)/z, \qquad A_t = 1 - t^2.$$

Reverse triangle gives $-1 \leq t \leq 1$. Squaring the defining identity gives

$$\frac{\mathrm{Re}\, h}{s} = -zt - z^2A_t/2, \qquad \frac{(\mathrm{Im}\, h)^2}{s^2} = z^2A_t - z^3tA_t - z^4A_t^2/4.$$

Consequently

$$\frac{R_s(h)}{s} = z^3f(t) + z^4g(t), \qquad f(t) = \frac{t(1-t^2)}{2}, \quad g(t) = \frac{(1-t^2)^2}{8}. \tag{8.39}$$

Replacing t by $|t|$ only increases the right side. On $[0,1]$ put $a_0 = 1/\sqrt{3}$. Then

$$f(a_0) - f(t) = \frac{(t-a_0)^2(t+2a_0)}{2} \ge a_0(t-a_0)^2, \qquad g(a_0) = 1/18, \qquad |g'(t)| \le 1/(3\sqrt{3}).$$

The mean-value inequality followed by completing a square proves

$$R_s(h) \le \frac{|h|^3}{3\sqrt{3}s^2} + \left(\frac{1}{18} + \frac{Z}{36\sqrt{3}}\right)\frac{|h|^4}{s^3}, \qquad |h|/s \le Z. \tag{8.40}$$

At $h = 0$ the assertion is immediate.

Define the fixed mass coefficient $\zeta = (3/2)\sqrt{p_+} < 1$. For any nonnegative frozen pair of coefficients $F_{\mathrm{c}} \ge F_{\mathrm{s}}$ set

$$F_{\mathrm{eff}} = (1-p_-)F_{\mathrm{c}} + p_- F_{\mathrm{s}}, \quad F_{\mathrm{mix}} = \zeta F_{\mathrm{c}} + (1-\zeta)F_{\mathrm{s}}, \quad F_{\mathrm{pair}} = (3F_{\mathrm{c}} + F_{\mathrm{s}})/4. \tag{8.41}$$

Here and below, ordering is checked for the chosen majorants, not inferred from an ordering of the actual geometric quantities. The ordered cubic cross sum satisfies

$$\sum_{a(i)\ne a(j)} e_i^2 e_j \le \zeta\sqrt{N}X^3, \qquad \sum_{i,j} e_i^2 e_j \le \sqrt{N}X^3.$$

Indeed, if $x_a = \sqrt{\sum_{a(i)=a} e_i^2}/X$, then $\sum x_a^2 = 1$. The first sum is at most $\sqrt{Np_+}X^3(\sum x_a - \sum x_a^3)$. The inequalities $\sum x_a \le 2$ and $\sum x_a^3 \ge (\sum x_a)/4$ prove the claim. The latter follows from $4\sum x_a^3 \ge (\sum x_a^2)(\sum x_a)$; expanding gives a sum of $(x_a - x_b)(x_a^2 - x_b^2) \ge 0$. Also $\sum_{a(i)\ne a(j)} e_i^2 e_j^2 \le 3X^4/4$, since $\sum_a E_a^2 \ge X^4/4$. Pure powers are weighted by the number of cross partners, at most $N(1-p_-)$. These proofs use arbitrary column energies and do not assume constant perturbations in a cluster.

In particular the independent type bound used in (8.38) is $(L_t)_{\mathrm{mix}}$ with $L_t = m_t g_t/h_t$; take its minimum with L_{row} when $L_{\mathrm{c}} \ge L_{\mathrm{s}}$. For the scalar term of (8.31), put

$$v_t = \frac{g_t T}{2\rho}, \quad D_t = m_t + v_t, \quad Z_t = \frac{2D_t T}{\rho h_t}, \quad C_t = \frac{D_t^3}{3\sqrt{3}h_t^2}, \quad B_t = \frac{D_t^4}{h_t^3}\left(\frac{1}{18} + \frac{Z_t}{36\sqrt{3}}\right). \tag{8.42}$$

The argument of R_s has modulus at most $D_t(e_i/r_i + e_j/r_j)$, since $e_i e_j/(r_i r_j) \le T(e_i/r_i + e_j/r_j)/(2\rho)$. Keep $r_i r_j$ outside the scalar remainder until expanding the powers:

$$r_i r_j(e_i/r_i + e_j/r_j)^3 = \frac{r_j}{r_i^2}e_i^3 + \frac{r_i}{r_j^2}e_j^3 + 3e_i^2 e_j/r_i + 3e_i e_j^2/r_j,$$
$$r_i r_j(e_i/r_i + e_j/r_j)^4 = \frac{r_j}{r_i^3}e_i^4 + \frac{r_i}{r_j^3}e_j^4 + 4e_i^3 e_j/r_i^2 + 4e_i e_j^3/r_j^2 + 6e_i^2 e_j^2/(r_i r_j).$$

Using (8.41) gives the full unsigned coefficients

$$\begin{aligned} A_3^u &= 2RC_{\mathrm{eff}}T/\rho^2 + 6C_{\mathrm{mix}}q_t/\rho, \\ A_4^u &= 2RB_{\mathrm{eff}}T^2/\rho^3 + 8B_{\mathrm{mix}}Tq_t/\rho^2 + 6B_{\mathrm{pair}}q_t^2/\rho^2, \\ A^u &= A_3^u + A_{\mathrm{mixed}} + A_4^u, \qquad B^u = B_{\mathrm{rad}}. \end{aligned} \tag{8.43}$$

The necessary C and B orderings are part of the certificate guards.

Lemma 8.9 (Shared-budget remainder)**.** *Suppose one complete scalar comparison supplies coefficients $A, B \ge 0$ such that the ordered remainder in (8.31), divided by Nd^2, is at most $A\vartheta^2 + B\vartheta\sqrt{1-\vartheta^2}$. Then it is at most*

$$\mathrm{Loss}(A,B) = \frac{A + \sqrt{A^2+B^2}}{2}. \tag{8.44}$$

In particular (8.43) supplies such a comparison.

Proof. The maximum is the larger eigenvalue of $\left(\begin{smallmatrix} A & B/2 \\ B/2 & 0 \end{smallmatrix}\right)$. Equations (8.37), (8.38) and the preceding power expansions establish the unsigned assertion with the same ϑ in every term. □

8.10 A signed remainder bound

Let $P_s(h) = -(\operatorname{Re} h)(\operatorname{Im} h)^2/(2s^2)$ be the scalar cubic. Substitution in (8.39) yields the exact identity

$$\frac{R_s(h) - P_s(h)}{s} = z^4 H_z(t), \qquad H_z(t) = \frac{(1-t^2)(5t^2-1)}{8} + \frac{3z}{8}t(1-t^2)^2 + \frac{z^2}{16}(1-t^2)^3. \tag{8.45}$$

This identity includes every order in h. Replacing t by $|t|$ raises its right side, and $H_z(t)$ is nondecreasing in z for $0 \le t \le 1$. More generally choose a fixed sign weight $\omega \in [0,1]$ and a rational $c_3 \ge 1/(3\sqrt{3})$. For $Z > 0$ let

$$W_{Z,\omega}(t) = \frac{(1-\omega)(f(t)-c_3)}{Z} + (1-\omega)g(t) + \omega H_Z(t).$$

If $Q_\omega(Z) \ge \max_{0\le t\le 1} W_{Z,\omega}(t)$, then

$$R_s(h) \le \omega P_s(h) + (1-\omega)c_3|h|^3/s^2 + Q_\omega(Z)|h|^4/s^3, \quad |h|/s \le Z. \tag{8.46}$$

To verify it, subtract the first two terms, divide by sz^4, and use (8.45). The result is $(1-\omega)(f(t)-c_3)/z + (1-\omega)g(t) + \omega H_z(t)$. Since $f(t) \le c_3$, replacing $0 < z \le Z$ by Z increases it. The zero case is direct. No condition $Z < 1$ enters this proof.

The continuum maximum is bounded by rational Bernstein coefficients. The ascending power coefficients of H_Z are

$$\left(\frac{-2+Z^2}{16}, \frac{3Z}{8}, \frac{12-3Z^2}{16}, -\frac{3Z}{4}, \frac{-10+3Z^2}{16}, \frac{3Z}{8}, -\frac{Z^2}{16}\right).$$

Add the displayed f, g, c_3 terms to obtain $W_{Z,\omega}$. On a closed interval $[a,b] \subset [0,1]$, substitute $t = a + (b-a)v$ and call the resulting power coefficients c_j. Its degree-six Bernstein coefficients are

$$\beta_i = \sum_{j=0}^{i} c_j \frac{\binom{i}{j}}{\binom{6}{j}}, \qquad 0 \le i \le 6.$$

The identity $W_{Z,\omega}(a+(b-a)v) = \sum_i \beta_i \binom{6}{i} v^i (1-v)^{6-i}$ and the nonnegative partition of unity bound the whole interval by $\max_i \beta_i$. The certificate uses the 32 closed equal intervals. It checks $Q_\omega(Z) \ge 0$ before substituting the nonnegative power majorants in the ordered sums.

We next explain the pure-row cubic input; its sharpened version is proved in the following subsection. For a fixed column j choose a unit vector $v_j \perp z_j$ and set

$$\ell_{ij} = \overline{S}_{ij} z_i^* K v_j, \qquad q_{ij} = |\ell_{ij}|^2/s_{ij}^2.$$

Stationarity, with its full stress term, is

$$\frac{1}{N}\sum_i \ell_{ij} = \lambda z_j^* K v_j, \qquad |\lambda z_j^* K v_j| \le \lambda_+ \min\{\kappa r_+, m_{\mathrm{s}}\}. \tag{8.47}$$

Its right side is generally nonzero. The Fourier identity

$$P_s(\ell u) = \operatorname{Re}\{\ell^3 u^3 - |\ell|^2 \ell |u|^2 u\}/(8s^2)$$

shows that a pure row bound follows from its first and third harmonics. Given $q_{\mathrm{c}} \in [q_-, q_+]$, $q_{\mathrm{s}} \in [0, q_{\mathrm{s},+}]$ and any fixed $c \ge 0$, define

$$\begin{aligned} F_1(c) = {} & c\lambda_+ \min\{\kappa r_+, m_{\mathrm{s}}\} + (1-p_-)\max\{|q_- - c|, |q_+ - c|\} m_{\mathrm{c}} \\ & + p_+ \max\{c, |q_{\mathrm{s},+} - c|\} m_{\mathrm{s}}, \end{aligned}$$

$$F_3 = (1-p_-)m_{\mathrm{c}}^3/h_{\mathrm{c}}^2 + p_+ m_{\mathrm{s}}^3/h_{\mathrm{s}}^2, \qquad P_{\mathrm{row}} = (F_1(c) + F_3)/8. \tag{8.48}$$

Subtract c times (8.47) from the first harmonic and apply the triangle inequality; the third harmonic uses its full typed triangle bound. Thus

$$N^{-1}\sum_i P_{s_{ij}}(\alpha_{ij}) \le P_{\mathrm{row}} e_j^3. \tag{8.49}$$

Valid generic ratio intervals are

$$\begin{aligned} q_- &= \max\left\{0, \frac{(1-\kappa)^2 r_-^2}{(1/\sqrt{3}+d_g)^2} - r_-^{-2}\right\}, \\ q_+ &= \min\left\{\frac{m_{\mathrm{c}}^2}{h_{\mathrm{c}}^2}, \frac{(1+\kappa)^2 r_+^2}{h_{\mathrm{c}}^2} - r_+^{-2}\right\}, \qquad q_{\mathrm{s},+} = m_{\mathrm{s}}^2/h_{\mathrm{s}}^2. \end{aligned}$$

They follow from the exact physical projection identity proved below. The certificate can intersect these intervals with sharper valid intervals and choose $c = 2$ or their endpoint midpoint. Its chosen c is fixed for the infinite-range argument.

To retain all mixed cubics, the symmetric trilinear form of P_s obeys $|\mathcal{T}_s(u,u,v)| \le |u|^2|v|/(3\sqrt{3}s^2)$. For $u = x + iy$, the numerator norm is $\sqrt{y^4 + 4x^2y^2}/6$, whose maximum at fixed $|u|$ gives this bound. Expansion of the cubic, or integration of its gradient, gives

$$P_s(u+v+w) \le P_s(u) + P_s(v) + \frac{(|u|+|v|+|w|)^3 - |u|^3 - |v|^3}{3\sqrt{3}s^2}.$$

All coefficients of the last polynomial are nonnegative. For each pair type retain v_t, D_t, Z_t from (8.42) and set

$$\begin{aligned} C_{\mathrm{base},t} &= c_3 m_t^3/h_t^2, \\ C_{\mathrm{extra},t} &= c_3(3m_t^2 v_t + 3m_t v_t^2 + v_t^3)/h_t^2 + (1-\omega)C_{\mathrm{base},t}, \\ C_{\mathrm{full},t} &= c_3(m_t+v_t)^3/h_t^2, \qquad B_{\mathrm{signed},t} = D_t^4 Q_\omega(Z_t)/h_t^3. \end{aligned} \tag{8.50}$$

In particular C_{full} does not receive an additional $(1-\omega)C_{\mathrm{base}}$ term. Outward rounding may increase D_t, Z_t; the scalar certificate then uses that rounded radius.

The physical radial factors require a separate estimate. The two pure ordered cubic terms combine to $2\sum_j r_j^{-2}\sum_i r_i P_{s_{ij}}(\alpha_{ij})$. One cannot pull a positive upper bound for r_i out of this signed sum. Write $r_i = 1 + t_i$ and apply (8.49) only to the part with coefficient one. With

$$C_{\mathrm{rms}} = \sqrt{(1-p_-)C_{\mathrm{base,c}}^2 + p_- C_{\mathrm{base,s}}^2}, \qquad B_{\mathrm{weight}} = \frac{2\omega C_{\mathrm{rms}} T q_d}{\rho^2 r_-},$$

Cauchy–Schwarz bounds the remaining part by $B_{\mathrm{weight}}\vartheta^2\sqrt{1-\vartheta^2}$ after division by Nd^2. Indeed $\sum_i |t_i| C_{\mathrm{base},t(i,j)} \le \sqrt{N} C_{\mathrm{rms}} T_r$ and $\sum_j e_j^3 \le TX^2$. It is safe to replace ϑ^2 by ϑ here.

Using (8.41), the complete signed alternative is

$$\begin{aligned} A_3^s &= \frac{2\omega P_{\mathrm{row}} T}{\rho^2} + \frac{2RC_{\mathrm{extra,eff}}T}{\rho^2} + \frac{6C_{\mathrm{full,mix}}q_t}{\rho}, \\ A_4^s &= \frac{2RB_{\mathrm{signed,eff}}T^2}{\rho^3} + \frac{8B_{\mathrm{signed,mix}}Tq_t}{\rho^2} + \frac{6B_{\mathrm{signed,pair}}q_t^2}{\rho^2}, \\ A^s &= A_3^s + A_{\mathrm{mixed}} + A_4^s, \qquad B^s = B_{\mathrm{rad}} + B_{\mathrm{weight}}. \end{aligned} \tag{8.51}$$

The certificate checks cross/same ordering for C_{base}, C_{extra}, C_{full} and B_{signed}, and nonnegative quartic coefficients. Lemma 8.9 then applies. Taking the smaller of complete unsigned and complete signed bounds, including the four fixed choices $\omega = 1/4, 1/2, 3/4, 1$, is valid. Their cubic and scalar-tail pieces are not selected independently.

8.11 Exact two-pair fibers and row cancellation

Here the first index of a fiber denotes its tangent center. Retain $\mathcal{T}, d_a, \ell_a$ from (8.16), and put

$$t_{2a} = d_a^2 + \sum_{b\neq a} \ell_a^2 \ell_b^2 p_a p_b, \qquad W_a = \frac{2t_{2a}}{p_a(\operatorname{tr}\mathcal{T})^2}, \qquad R_a = \frac{3\lambda - 2 + 2p_a}{6p_a(2\lambda-1)}. \tag{8.52}$$

Then $W_a = \|Kz_a\|^2$ and $R_a = \|z_a\|^2$. Indeed $\mathcal{T}UU^*\mathcal{T} = \mathcal{T}^2$ since UU^* projects onto its range; its diagonal gives the formula for W_a. The formula for R_a is the candidate diagonal identity $p_aR_a = 1/4 + (p_a - 1/4)/[3(2\lambda-1)]$. Complex dimension two implies that for every unit $v \perp z_a$,

$$m_{ab}^2 := |z_b^* K v|^2 = W_b - s_{ab}^2/R_a, \qquad q_{ab} = m_{ab}^2/s_{ab}^2. \tag{8.53}$$

This follows by subtracting the projection of Kz_b on $z_a/\|z_a\|$. In particular the modulus is independent of the choice of unit vector in this one-dimensional complex fiber, without removing its phase. For $a = b$ the overlap is $s_{aa} = d_a/(p_a \operatorname{tr}\mathcal{T})$, not the cross formula $\ell_a\ell_b/\operatorname{tr}\mathcal{T}$.

For the profile $C = 1$ put

$$\begin{gathered} x = 1/N, \quad t = \sqrt{3-8x^2}, \quad a = \tfrac{1}{4} - \tfrac{x}{2}, \quad b = \tfrac{1}{4} + \tfrac{x}{2}, \\ (p_1,p_2,p_3,p_4) = (a,a,b,b), \quad \lambda = (1+t)/4. \end{gathered} \tag{8.54}$$

Substitution in (6.1), using $t^2 + 8x^2 = 3$, proves the last identity by the uniqueness of the root in $(2/3, \alpha]$. Reversing the signed count profile permutes the two pairs. There are six ordered types: $\mathrm{aa}, \mathrm{ab}, \mathrm{ba}, \mathrm{bb}, \mathrm{same}_a, \mathrm{same}_b$. Here aa means two different clusters of mass a and must not be confused with same_a. The following explicit formulas make their interval verification independent of a choice of spinor phases:

$$s_{\mathrm{aa}} = \frac{t(2x+1)}{(2x-1)(8x^2-3)}, \qquad s_{\mathrm{ab}} = \frac{-t}{8x^2-3}, \qquad s_{\mathrm{same}_a} = -\frac{2tx + 8x^2 - 3}{(2x-1)(8x^2-3)}. \tag{8.55}$$

Set $V = 4tx + 20x^2 + 8x - 3$ and $J = -32tx^3 - 12tx^2 + 6tx + 16x^4 + 56x^3 - 36x^2 - 24x + 9$. Then

$$\begin{aligned} m_{\mathrm{aa}}^2 &= \frac{-2V}{(2x-1)(44x^2+12x-9)}, \\ m_{\mathrm{ab}}^2 &= \frac{2J}{(2x+1)(8x^2-3)(44x^2+12x-9)}, \\ m_{\mathrm{same}_a}^2 &= \frac{8x^2V}{(2x-1)(8x^2-3)(44x^2+12x-9)}, \\ q_{\mathrm{aa}} &= \frac{2(2x-1)(8x^2-3)V}{(2x+1)^2(44x^2+12x-9)}, \\ q_{\mathrm{ab}} &= \frac{-2J}{(2x+1)(44x^2+12x-9)}, \\ q_{\mathrm{same}_a} &= \frac{8x^2(-32tx^2+104x^3-36x^2-42x+9)}{9(2x-1)(2x+1)(44x^2+12x-9)}. \end{aligned} \tag{8.56, 8.57}$$

All formulas for center b are obtained by replacing x by $-x$ and interchanging the labels a, b. These identities follow by substitution in (8.52) and (8.53), followed by multiplication by the denominators and reduction using $t^2 = 3 - 8x^2$. This supplies direct polynomial checks of each identity. The factor x^2 in both same-type formulas is retained before interval evaluation.

For a center of mass a, denote the four complex forms in (8.47) by ℓ_0 in its own cluster, ℓ_1 in the other a cluster, and ℓ_2, ℓ_3 in the b clusters. Exact stationarity reads

$$a\ell_0 + a\ell_1 + b\ell_2 + b\ell_3 = \lambda\ell_0.$$

The two b ratios are the same real number q_{ab} even though their complex forms need not have the same phase. Thus the first harmonic has the exact reduction

$$\sum_j p_j q_j \ell_j = a(q_{\mathrm{aa}} - q_{\mathrm{ab}})\ell_1 + [a q_{\mathrm{same}_a} + q_{\mathrm{ab}}(\lambda - a)]\ell_0. \tag{8.58}$$

There is no assertion that the individual b terms vanish. Since $\lambda > a$,

$$\begin{aligned} F_{1a} &= a|q_{\mathrm{aa}} - q_{\mathrm{ab}}|m_{\mathrm{aa}} + [a q_{\mathrm{same}_a} + q_{\mathrm{ab}}(\lambda - a)]m_{\mathrm{same}_a}, \\ F_{3a} &= a q_{\mathrm{same}_a} m_{\mathrm{same}_a} + a q_{\mathrm{aa}} m_{\mathrm{aa}} + 2b q_{\mathrm{ab}} m_{\mathrm{ab}}, \\ P_{\mathrm{row}}^{\mathrm{pair}} &= \max\{F_{1a} + F_{3a}, F_{1b} + F_{3b}\}/8 \end{aligned} \tag{8.59}$$

is a valid bound in (8.49). The third harmonic uses $|\ell_j|^3/s_j^2 = q_j m_j$ exactly. The b center uses the exchanged formulas. In particular the same-cluster stress term survives, and all physical residual vectors are eligible. One may take the minimum with (8.48).

For clarity, the finite continuum certificate for these formulas uses the 512 closed intervals $[j/(180\cdot 512), (j+1)/(180\cdot 512)]$, $0 \le j < 512$. On each interval $[u, v]$ use $\sqrt{3 - 8v^2} \le t \le \sqrt{3 - 8u^2}$ with directed rational square roots. Rational interval substitution in (8.55)–(8.57), for all six types, checks that every denominator excludes zero. Geometric nonnegativity permits clipping a lower bound for m^2, q to zero; a negative upper bound fails the check. Evaluate (8.59) on each same interval, retaining both centers. The complete recorded bounds on $0 \le x \le 1/180$ include

$$\begin{aligned} m_{\mathrm{c}}^2 &\le \frac{674866483913819}{10^{15}}, & m_{\mathrm{s}}^2 &\le \frac{1730019549}{62500000000000}, \\ \frac{236851236625637}{125000000000000} \le q_{\mathrm{c}} &\le \frac{2109213874049103}{10^{15}}, & q_{\mathrm{s}} &\le \frac{13969617457}{500000000000000}, \\ P_{\mathrm{row}}^{\mathrm{pair}} &\le \frac{160219816186209}{10^{15}}. \end{aligned} \tag{8.60}$$

At the local endpoint 182, every whole interval intersecting $[0, 1/182]$ is retained, including a crossing boundary interval. The slightly sharper pure-row cap used there is $32029714476747/200000000000000$. This is an interval cover, not a maximum over sampled configurations. Only the pure-row term is replaced in (8.51); B_{weight} and every other remainder term are still present.

8.12 Propagation and equality

We now extend the local comparison from a certified endpoint to all later integers of the same profile. On each closed localization region, fix its $U, W, L_U, L_P, \bar{E}$ and the simultaneous candidate coefficients B, C_{pt} supplied by Sections 7 and 6. At an accepted endpoint n, put $e_n = m_*/n$ and $\sigma_n^2 = C/n^2$. Let κ_* be the outward rational upper enclosure of n times the explicit stress bound in (6.23):

$$\kappa_* \ge 2\sqrt{C}\sqrt{1/3 + 4e_n/b_0 + 4\sigma_n^2/b_0^2} + \frac{2ne_n^2}{1/15 - e_n}, \qquad b_0 = 109/100.$$

This coefficient is distinct from the Bloch-Gram coefficient k_0 in Section 6. We retain that accepted enclosure throughout propagation. At this endpoint, valid physical coefficients are

$$s_n = \sqrt{13U/(4n)}, \qquad A_R = \frac{13U}{4} + L_U W \frac{1+s_n}{(1-s_n)^2}, \qquad P_R = \frac{13U}{4} + L_P W \frac{1+s_n}{(1-s_n)^2}. \tag{8.61}$$

These follow from the radial/angular identity, with the relative angular bound $L_U w/[N(1-s)^2]$ and its pointwise analogue. Their minimum with an independently certified bound in the

original variables is also valid. The partition and the closed count cover remain fixed, so the count exclusions apply at every later local endpoint.

At an accepted endpoint, all of the following are checked by outward rational inequalities: positive masses and overlap floors; the candidate point domains; every fine phase guard and the separate closest-gauge rotation guard; the mean-coordinate phase, projection, and positivity guards; the intrinsic residual lower bound; positive radial multipliers; both finite-normal branches; the nonnegative quartic and cross/same ordering guards; and the full remainder margin. The square root of a positive rational is enclosed by rational numbers whose squares straddle it, so these checks require no floating-point eigenvalue or optimizer. Section 9 specifies the interval records, arithmetic and source binding.

Lemma 8.10 (Frozen endpoint propagation)**.** *Suppose all stated guards and a strict complete local margin hold on one closed region at endpoint n, with intrinsic and exact-fiber enclosures valid on $0 \le 1/N \le 1/n$. Then the same margin lower bound holds on that region for every later admissible member of its profile.*

Proof. For $N \ge n$, retain the fixed profile deviations δ_a and use the actual masses $p_a = 1/4 + \delta_a/N$. The candidate is reconstructed at those masses; it is not obtained by scaling a finite-endpoint vector configuration. Its errors obey

$$e = m_*/N, \qquad \eta \le \sqrt{C_{\mathrm{pt}} C}/N, \qquad \kappa \le \kappa_*/N, \qquad \tau \le (2/3)C/N^2.$$

For the fixed count profile, the stress estimate follows because N times the explicit bound in (6.23) is nondecreasing in $e = m_*/N$ and $\sigma^2 = C/N^2$; its value for $N \ge n$ is therefore bounded by the accepted enclosure κ_*. Freeze every favorable lower coefficient and every unfavorable upper coefficient of the accepted endpoint. Here is the justification for each part of this operation.

First, the global exclusions remain strict and the regional fine bounds remain valid by Section 7. In (8.61), s_n and $(1+s_n)/(1-s_n)^2$ decrease as the endpoint increases. The candidate errors decrease, r_- increases, and r_+ decreases. Thus all angular transport majorants, T, q_t, q_d and R have their endpoint upper bounds; ρ has its endpoint lower bound. Both actual-rotation guards remain valid. No difference of decreasing functions is asserted to be monotone.

Second, each nonnegative perturbation, phase-defect and projection majorant in the mass and stress mean estimates decreases when these candidate errors decrease. Their favorable metric factors increase. Hence the accepted physical mean lower bound remains valid. The intrinsic c_K was certified on the entire reciprocal-N domain, and multiplication by $1-\kappa$ preserves its endpoint lower bound. The original physical normal estimates similarly retain their fixed upper J, L. The normal radius $d_c/(\sqrt{2}N)$ shrinks. Therefore the supremum in (8.23) cannot increase: its coefficient bounds are frozen and its domain only shrinks. This argument does not keep a particular active maximizing branch.

Third, all generic projected and phase-row majorants have the displayed endpoint bounds. The exact two-pair inequalities cover the whole reciprocal domain, without needing monotonicity of an individual ratio. Freeze λ_+ and c_K separately in the spectral row bound; then $\lambda_+(1+\kappa) - (1-\kappa)c_K$ has its endpoint upper bound. The cross-only row shares this bound, while the independent pair coefficient M is retained.

Finally freeze every scalar radius cap Z_t, sign weight ω, Bernstein upper bound $Q_\omega(Z_t)$, harmonic center c, and the separate positive cross/same type coefficients. All later increments lie in the same certified disks. The nonnegative type polynomials and the signed physical-weight estimate give the endpoint upper coefficients in (8.51); the unsigned bound propagates in the same way. The ordering checks concern these fixed majorants. No monotonicity of a rounded Bernstein output or of a selected numerical branch is needed. □

The rational certificates in Section 9.3 cover all 7320 budget cells by strict count exclusions and 228 accepted closed local regions. On every accepted region, the certified inequality (9.2) gives $g_{\mathrm{normal}} - \mathrm{Loss} > 1/10000$. The detailed coverage totals and rational margins are recorded there.

Proposition 8.11 (Uniform strict local comparison)**.** *For every nondivisible integer $N \geq 180$ and every localized competitor with its actual closest candidate and gauge,*

$$\Gamma_K(Y) \leq N^2\lambda - \frac{N}{10000}\|Y - Z\|_F^2. \tag{8.62}$$

Proof. If $d = 0$, stationarity gives equality in (8.62) directly. Otherwise, since $D\mathcal{L}(Z) = 0$, the second-order Taylor polynomial of $\mathcal{L}(Z + \Delta) - \mathcal{L}(Z)$ is $Q(\Delta)$. The quadratic estimate is at most $-Ng_{\mathrm{normal}}d^2$, and the ordered Taylor remainder is at most $N\,\mathrm{Loss}\,d^2$. Every retained closed region has $g_{\mathrm{normal}} - \mathrm{Loss} > 1/10000$ by the rational checks just stated. The preceding propagation argument proves the entire subsequent range. Odd integers begin at 181; residue-two integers begin at 182, so the local endpoint 182 for $C = 1$ leaves no integer gap. □

The comparison applies to every localized kernel, with no restriction on the number of distinct column directions. In the ordered global trace chain, equality in the optimal bound therefore forces $d = 0$ by Proposition 8.11. Strict integer balancing then forces the balanced profile, and every nonzero candidate stress overlap fixes its corresponding matrix entry. This gives the unitary, individual-phase, permutation and conjugation classification in the main theorem. The divisible integers, including 180, are handled by the exact divisible case of the global theorem. Hence 180 is a sufficient threshold.

9 Exact certificates and the computational proof

The computer-assisted part of the proof consists of finite rational inequalities on closed domains. We specify the domain covers, the acceptance conditions and their propagation to infinitely many integers. The accompanying reproduction package contains the fixed certificates and an arithmetic verifier that checks them without repeating the search.

9.1 Arithmetic and acceptance rules

All arithmetic used for acceptance is rational. To enclose $\sqrt{a/b}$, for nonnegative integers a and positive b, choose a positive integer scale S and compute

$$q = \left\lfloor \sqrt{\lfloor aS^2/b \rfloor} \right\rfloor.$$

The lower endpoint is q/S. If $bq^2 = aS^2$, the upper endpoint is also q/S; otherwise it is $(q+1)/S$, strictly above $\sqrt{a/b}$. Interval addition, multiplication, reciprocal and square roots are rounded outward. Division is accepted only after its denominator interval excludes zero, and every phase, overlap and covariance guard is checked before use. Fixed decimal grids are rational denominator grids, not floating-point error tolerances. The stored scales and exact endpoints are part of the certificates.

A scalar strict inequality is accepted only when the rational upper bound is strictly below the rational lower bound on the other side. We use $1/10000$ as a common certified lower bound for the local margin, independently of any auxiliary target used to refine an interval cover. In particular, the continuous residual records marked `FAIL_TARGET` retain rigorous positive lower bounds. The local proof uses these bounds, regardless of whether they meet the auxiliary target.

9.2 Global and continuous-parameter certificates

The anchor potential (7.7) uses 256 closed intervals per budget. With $K_A = 441/100$ and $\eta_A = 8/25$, its saved upper maximum is

$$\frac{17017848239490016205904994700214316636180000000000000000}{17030198754513787096073220152288809287941323918941181 39} < 1.$$

The same records provide the strict coverage, disjointness and scatter inequalities, and the separate count checks for $M = 1/2$ and $11/19$. The fine chain (7.12) has 9 recorded stages, each with 128 closed budget bins. Its 38 explicit guards include all positive denominators and forward implications. Regional fine records then retain their own 64-bin bounds and phase choices.

The candidate spectra and overlap floors cover $0 \le x = 1/N \le 1/180$. The cover is the union of the saved $[0, 1/200]$ domain and the closed strip $[1/200, 1/180]$. The latter has 2048 overlap boxes and 16245 residual leaves, with all four profiles and both signs. The shared endpoint is present in both domains. A later endpoint n uses whole saved cells meeting $[0, 1/n]$, including the whole of a boundary-crossing cell. Its extra portion only weakens the bound.

The root enclosures follow from the scalar equation. Put $h_0 = \alpha - 1/2$, $\tau = \alpha - \lambda$ and $h = h_0 - \tau$. Exact division of the scalar root equation gives

$$\tau = \frac{x^2 W(h)}{D(h)} \sum_a \frac{\delta_a^2}{h - 2\delta_a x}, \qquad W(h) = 3 + 1/h - 1/(4h^2), \quad D(h) = 2 + 1/(4h_0 h). \tag{9.1}$$

Start with $0 \le \tau \le (2/3)Cx_{\mathrm{hi}}^2$ and intersect this interval with the interval evaluation of (9.1) three times. Existence and uniqueness of the root have already been proved analytically. Each intersection contains that root because it satisfies the displayed identity; no contraction factor or convergence assumption is needed. Every before, right-side and after interval is stored, and an empty intersection is a failure.

For the balanced residue-two profile, the sharper formula $\lambda = (1 + \sqrt{3 - 8x^2})/4$ is used for the six ordered projected fiber types in Section 8. The exact rational expressions are enclosed on 512 closed intervals in $[0, 1/180]$. The stationary row certificate reads these intervals and bounds both possible center types on the same bins. Sign reversal interchanges the two mass types. The domain covers justify the physical m, q and P_{row} bounds throughout the stated range, without a monotonicity assumption.

9.3 Closed budget covers and local comparisons

For each of the four count profiles divide the u, w triangle into the closed cells

$$\frac{iM}{60} \le u \le \frac{(i+1)M}{60}, \quad \frac{jM}{60} \le w \le \frac{(j+1)M}{60}, \quad u + w \le M, \qquad i, j \ge 0,\ i + j \le 59.$$

There are $60 \cdot 61/2 = 1830$ cells per profile. Boundary overlaps are intentional. On a cell, $v \le M - u_{\mathrm{lo}} - w_{\mathrm{lo}}$, and (7.8) with the valid seed 31 supplies a rational upper bound for $\sqrt{C}$. A cell is excluded only if this is strictly below the profile's $\sqrt{C}$ lower enclosure.

In each radial row, descend through angular indices until the highest nonexcluded cell is reached. Its local comparison is evaluated on the *enlarged closed region* $0 \le u \le u_{\mathrm{hi}}$, $0 \le w \le w_{\mathrm{hi}}$, $u + v + w \le M$. Consequently one comparison covers all lower angular cells in that row. This does not assume that coefficients computed on different regions are monotone. Fully excluded rows retain all their exclusion inequalities. The complete accounting is

C	Local endpoint	Excluded	Covered by regions	Regions
3/4	180	42	1788	60
11/4	180	1027	803	53
3	180	908	922	55
1	182	62	1768	60
Total		2039	5281	228

The odd-profile minimum margins are, respectively,

$$\frac{537329859213}{40000000000000} \quad \text{and} \quad \frac{3713814057647}{10^{15}}.$$

The $C = 3$ exact margin is a longer rational stored in its certificate; exact cross multiplication proves it exceeds $1/10000$ (its decimal value is approximately 0.0001017243329754284). For $C = 1$ the final lower margin is

$$\frac{1420911820069}{500000000000000} > \frac{1}{10000}.$$

Thus the uniform margin in (4.6) is valid.

For $C = 3$ we combine the stress-coordinate mean estimate and finite-normal maximization with the unsigned remainder. For $C = 1$ we use the candidate, localization and normal bounds at endpoint 182, the spectral row estimate and the partial-signed remainder bounds. The four sign weights are $1/4, 1/2, 3/4, 1$. Each row retains all complete alternatives and selects a valid minimum, with positive quartic factors and checked cross/same coefficient ordering. There are 480 stored scalar tail certificates, each covering 32 closed intervals of the degree-six Bernstein polynomial. These bounds include the mixed terms and the physical radial-weight correction.

The finite verification statement used in Proposition 4.4 is as follows. For each of the 228 enlarged regions, independently reconstruct the candidate bounds, localization interface, physical quadratic coefficients c_m, c_r, J, L, squared normal radius $r_n = \rho_n^2 = d_c^2/(2n^2)$, and one complete pair $A_{\mathrm{rem}}, B_{\mathrm{rem}}$. Acceptance requires all of the following:

1. The saved closed cells have precisely the declared endpoints, all signs, vertices and pair types are present, and the count exclusions and enlarged accepted regions cover all 7320 budget cells.

2. Every denominator, covariance, phase, rotation and overlap guard in Sections 7 and 8 holds. In particular $c_m, c_r > 0$, $J, L \geq 0$, $0 \leq r_n \leq 1/2$, $A_{\mathrm{rem}}, B_{\mathrm{rem}} \geq 0$; the radial floor is positive; and the required cross/same coefficient orderings hold.

3. The intrinsic and overlap inputs are enclosed on the whole joined reciprocal domain. In the refined $C = 1$ rows the spectral bound uses $\lambda_+ \geq (1 + \sqrt{3})/4$, which covers $x = 0$ as well as the finite endpoint. Every used signed scalar tail is bounded on its entire disk by the independently expanded degree-six Bernstein certificate.

4. Select the applicable branch of (8.24) using these conservative coefficients. With g_{normal} the negative of its upper enclosure and Loss^+ an upper enclosure of (8.44), exact rational comparison gives

$$g_{\mathrm{normal}} - \mathrm{Loss}^+ > \frac{1}{10000}. \tag{9.2}$$

The arithmetic verifier checks this statement on the fixed witnesses. Its analytic consequences and propagation are proved in Section 8.12.

9.4 The worst $C = 3$ region

The minimum over all 55 accepted $C = 3$ regions occurs at row $(i, j) = (17, 23)$. Its original cell has $u \in [187/1140, 33/190]$ and $w \in [253/1140, 22/95]$. The local inequality is certified on the larger set

$$0 \le u \le 33/190, \quad 0 \le w \le 22/95, \quad v \ge 0, \quad u + v + w \le 11/19, \tag{9.3}$$

at the local endpoint $n = 180$. Both signs of $\delta = (-3, 1, 1, 1)/2$ are included in the candidate certificates. The stress-coordinate mean bound and the physical normal coefficients are combined with the entire unsigned remainder, including the same-cluster, radial, mixed and quartic terms. The active normal branch is the interior branch. The following table gives rational enclosures for this calculation; the distributed `validation/C3_WORST_ACCEPTANCE.json` gives the exact local inputs, intermediate coefficients, recorded regional guards and the unabridged rational margin. Its source-report binding supplies the independently checked upstream candidate and localization interfaces.

Quantity	Direction	Rational bound
c_m	$\ge$	0.228928718452
c_r	$\ge$	0.210023739886
J	$\le$	0.815026271354
L	$\le$	0.803437886305
ρ_n^2	$\le$	0.000132756408
A_3^u	$\le$	0.156826320895
A_{mixed}	$\le$	0.018831369020
A_4^u	$\le$	0.014800634464
A^u	$\le$	0.190458324379
B^u	$\le$	0.109690746792
g_{normal}	$\ge$	0.205224509235
$\mathrm{Loss}(A^u, B^u)$	$\le$	0.205122784909
$g_{\mathrm{normal}} - \mathrm{Loss}$	$\ge$	0.000101724326

Every terminating decimal in this table denotes its exact rational value. For the displayed coefficients, exact cross multiplication gives

$$J^2\rho_n^2 \le (c_m - c_r)^2(1 - \rho_n^2), \qquad c_m - c_r > 0,$$

so the interior formula is applicable even after this additional outward rounding. The displayed A^u is the sum of its three displayed upper bounds. The gap and loss rows are recomputed from the displayed coefficients, rather than copied from a saved final margin. Their difference is strictly greater than $1/10000$. The unrounded certificate margin is approximately 0.0001017243329754. All 55 regions were compared as exact rationals to identify this minimum.

9.5 Propagation to all integers in the stated range

An accepted row fixes its entire region, count floors, fine coefficients, candidate domain, normal-gap bound, complete remainder alternative, sign weight and scalar Bernstein bounds. The term-by-term propagation argument in Section 8.12, concluding in Proposition 8.11, shows that the analytic error majorants retain their endpoint bounds and the permitted normal interval shrinks. The cross/same majorants remain fixed, so their ordering is preserved. A later comparison stays within the same accepted scalar disks and remainder bound. This argument applies throughout the subsequent range and does not depend on which branch a new evaluation would select.

The two odd profiles apply to all odd $N \geq 181$. The $C = 3$ profile applies from 180, and the $C = 1$ profile from 182; both are therefore valid for every residue-two integer in the theorem. The exact divisible case covers 180 and all later multiples of four. These branches give every integer $N \geq 180$. The local endpoints differ by profile; the resulting sufficient threshold need not be minimal.

9.6 Certificates and arithmetic verification

The certificates and verifier are publicly available at https://github.com/motives3/coherence-codimension-two-reproduction. This paper refers to the fixed snapshot at commit b7a51dbc324b. All `certificate/` and `validation/` paths in this paper are relative to the repository's `reproduction/` directory, which is the reproduction-package root. The archive `certificate/supplement.zip` contains the fourteen JSON records used by the arithmetic verifier. The machine-readable `certificate/INDEX.json` gives their archive-relative paths, byte sizes and SHA-256 digests, together with the archive digest. The guide `certificate/ARITHMETIC_README.txt` explains the verifier and its dependencies. The main adopted records are identified by the following short names:

Odd180 The complete subproofs for the two odd profiles in `PROFILE_180.json`, both with certified positive margins.

Stress180 The complete $C = 3$ subprofile of `STRESS_SAVED_NORMAL.json`.

Refined182 `REFINED_182_NORMAL.json`, bound to the immutable `FROZEN_FULL_182.json` input.

Continuous bounds The joined root-residual and weighted-overlap records. The two additional sharp records are `SHARP_FIBERS.json` and `SHARP_STATIONARY.json`.

The index supplies the full paths. Some records have an overall failure label but contain a complete subproof with a certified positive margin. The verifier checks the adopted subproofs identified above, including their domains and hypotheses; it does not use the status labels as acceptance criteria. The enclosure κ_* in Section 8.12 is stored in the certificate field `kappa_coefficient`; this field corresponds to the stress coefficient, not the Bloch-Gram coefficient k_0.

The verifier consists of four files and uses only the Python standard library:

```
certificate/verify_certificates.py
validation/global_arithmetic.py
validation/continuous_arithmetic.py
validation/local_arithmetic.py.
```

Run using Python 3.10 or later:

```
python -B certificate/verify_certificates.py
```

Run this command from the reproduction-package root. The program locates its data relative to its own file, reads the ZIP without extraction and emits JSON to standard output. It creates no files. A failed guard or arithmetic mismatch exits nonzero. Diagnostic partial stages are expressly marked incomplete.

The verifier recomputes the global and regional shared budgets, candidate polar bounds, fixed root intersections, intrinsic residual bounds, all overlap and projected-fiber intervals, stationary harmonic rows, and every Bernstein coefficient of the used scalar tails. It then reconstructs the physical mean/residual curvature, normal maximization, phase transport and complete local remainder on all 228 adopted regions. It does not accept a saved margin as an

input to strictness. The fixed covers and root boxes serve as witnesses, so verification requires no threshold search, adaptive subdivision or optimization. The accompanying regression sources exercise the arithmetic kernels and the rejection of corrupted witnesses.

The historical all-stage report `validation/ARITHMETIC_ACCEPTANCE.json` records the hashes of its fourteen input records and four verifier files. Their correspondence with the public distribution is documented in `validation/PRIOR_ACCEPTANCE_BINDING.json`; `PRIVACY_TRANSFORM.json` describes the metadata-only redactions. The current archive and input hashes are listed in `certificate/INDEX.json`.

The verifier checks finite rational inequalities. The analytic identities, geometric implications, infinite-range propagation and equality classification are proved in the manuscript; they are outside the verifier's scope. This is a computer-assisted proof, rather than a proof-assistant formalization.

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OpenAI's Codex was used extensively in the research and preparation of this manuscript, including use of GPT-6 Astra with the Ultra reasoning setting. Its use included literature searches and comparisons with earlier results (Section 2); exploration and development of the four-block construction and its algebraic formulas (Sections 3 and 6); and formulation, refinement, and critical examination of arguments concerning the matrix reduction, global localization, local comparison, and equality cases (Sections 4, 7, and 8).

Codex was also used in developing the computational work associated with Section 9 and the accompanying reproduction package. This included implementing and revising code for numerical exploration, certificate generation, rational-arithmetic verification, and regression tests.

AI-generated drafts of mathematical arguments and explanatory prose were used and revised during the preparation of the manuscript. Codex was also used to organize the presentation, revise notation and cross-references, and edit the English.

The author assumes responsibility for all content, including the mathematical claims, references, code, and final text.

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